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(54) **IMAGE-CAPTURING OPTICAL LENS ASSEMBLY**

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359/714

(58) **Field of Classification Search** 359/740,
359/753, 763, 770

See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

6,906,872 B2 * 6/2005 Wang et al. 359/753
7,446,955 B1 11/2008 Noda
2008/0144193 A1 * 6/2008 Adachi 359/770

* cited by examiner

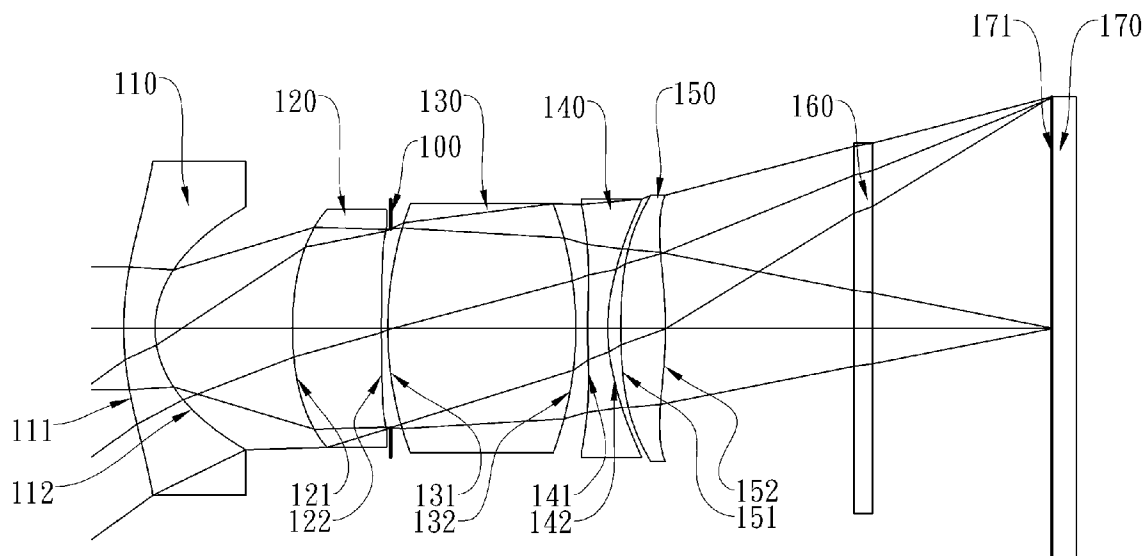
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(57) **ABSTRACT**

The present invention provides an image-capturing optical lens assembly comprising, in order from an object side to an image side, a front lens group, a stop, and a rear lens group. The front lens group comprises, in order from the object side to the image side: a first lens element with negative refractive power having a convex object-side surface and a concave image-side surface; and a second lens element with positive refractive power having a convex object-side surface and a concave image-side surface. The rear lens group comprises, in order from the object side to the image side: a third lens element with positive refractive power having a convex object-side surface and a convex image-side surface; a fourth lens element with negative refractive power; and a fifth lens element with positive refractive power having a convex image-side surface. With such an arrangement of optical elements, the optical system will have a field of view that is large enough; meanwhile, aberrations of the optical system can be favorably corrected to obtain good image quality.

18 Claims, 16 Drawing Sheets



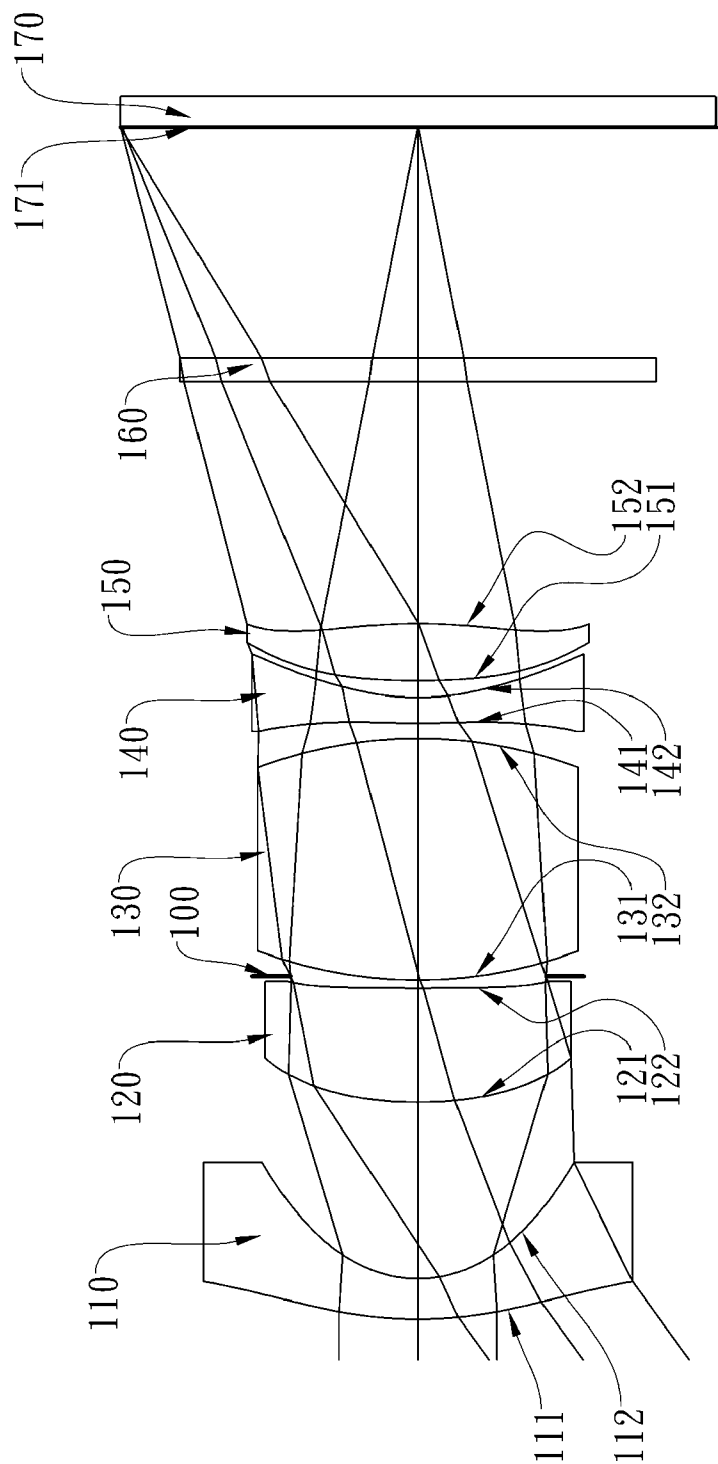


Fig. 1A

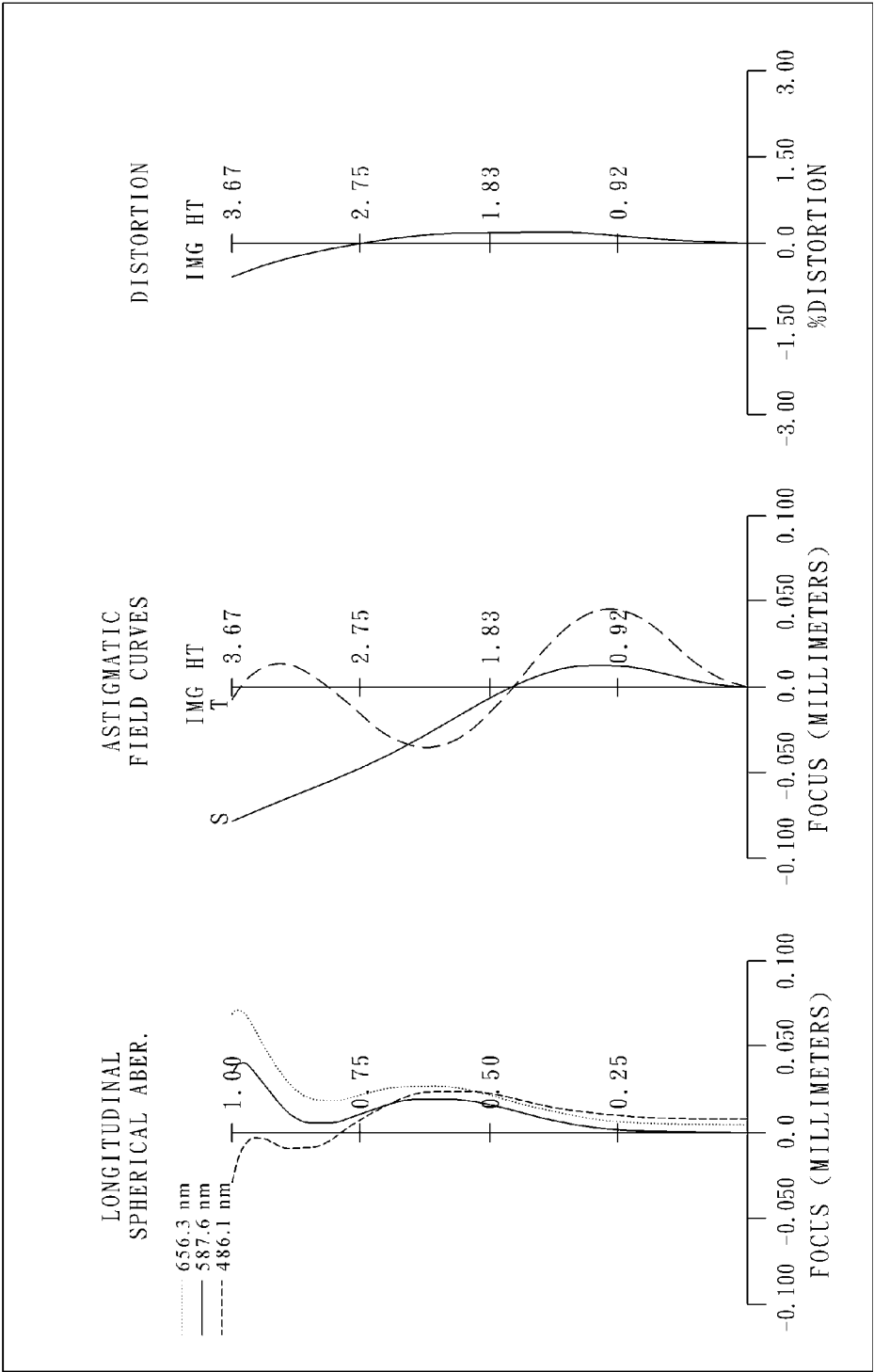


Fig. 1B

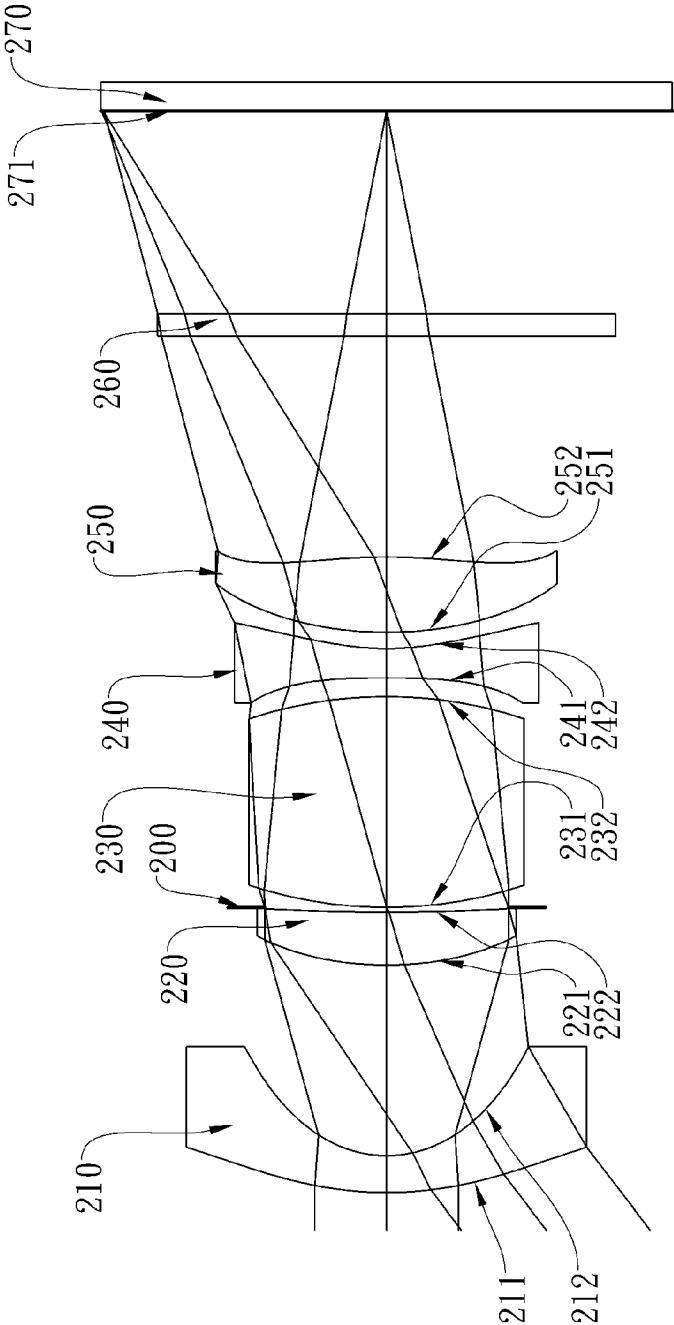


Fig. 2A

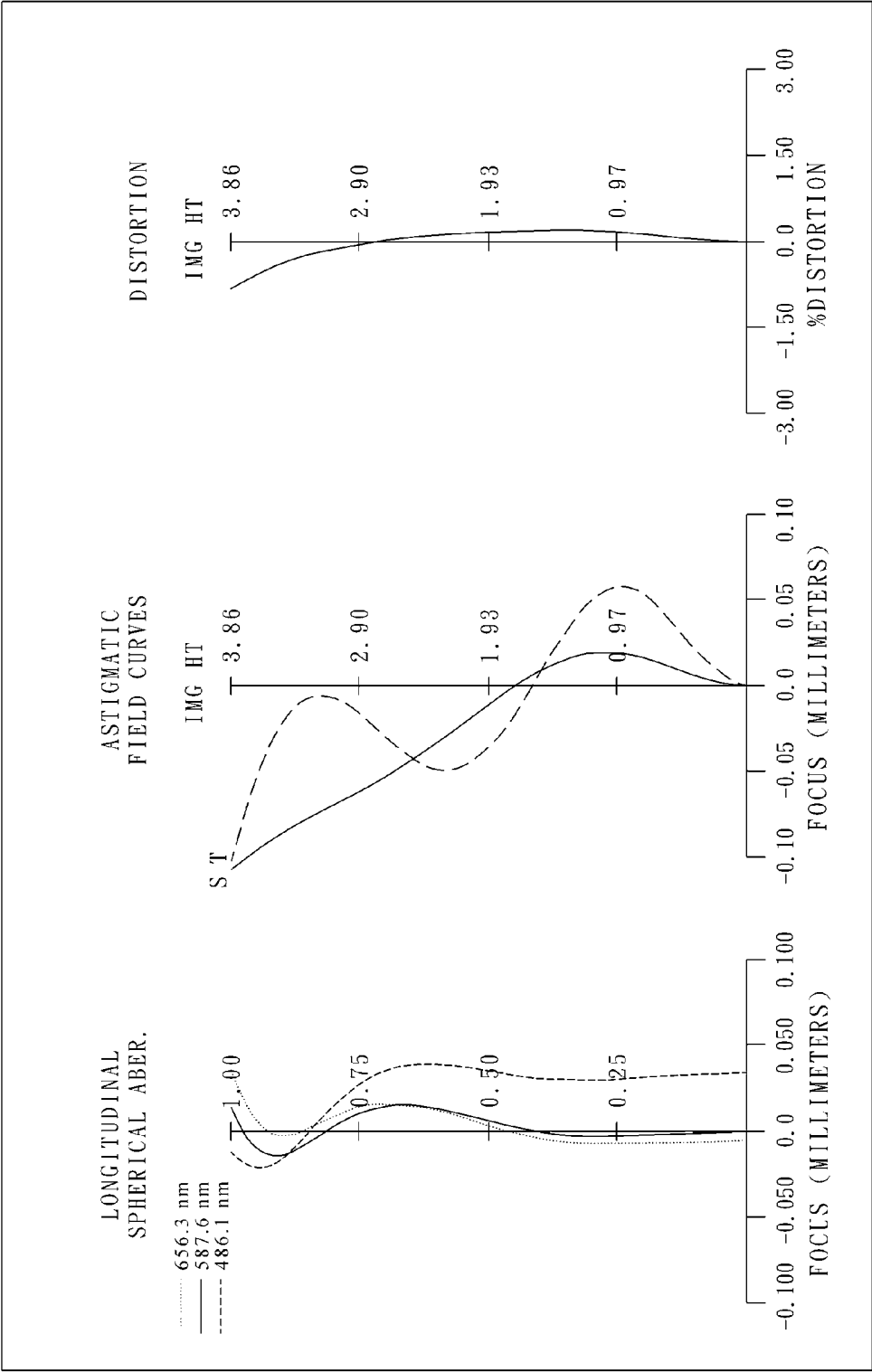


Fig. 2B

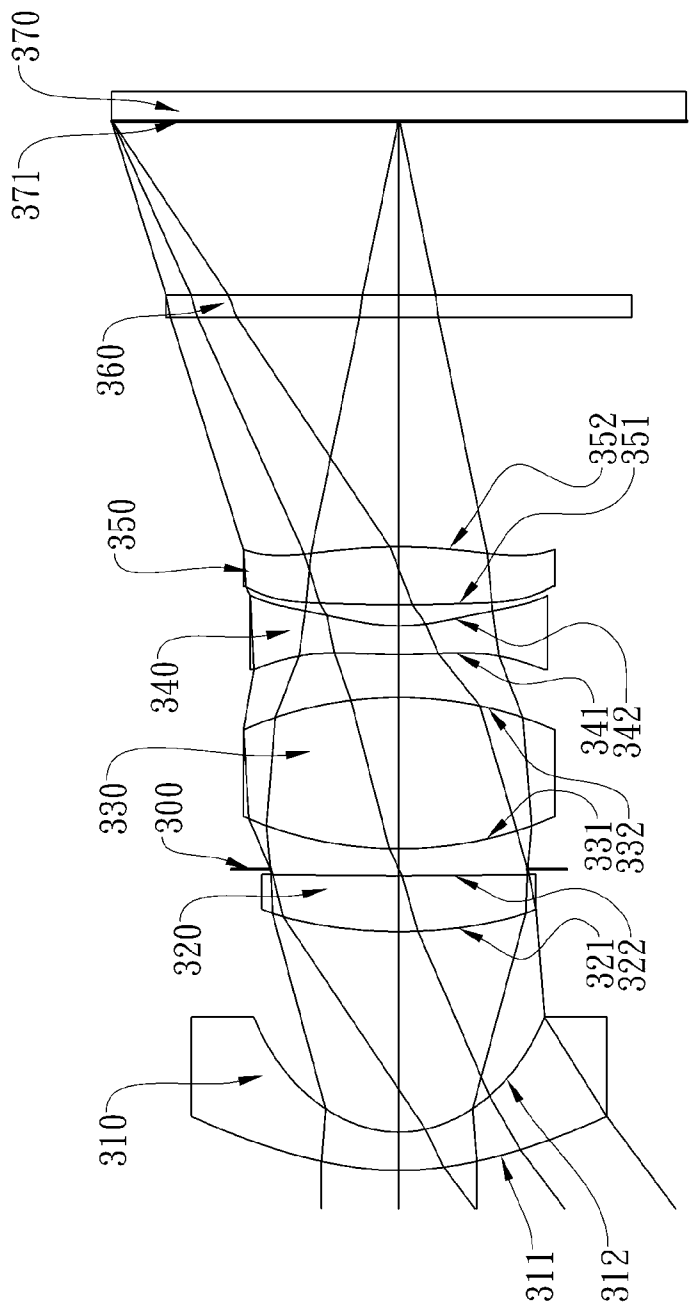


Fig. 3A

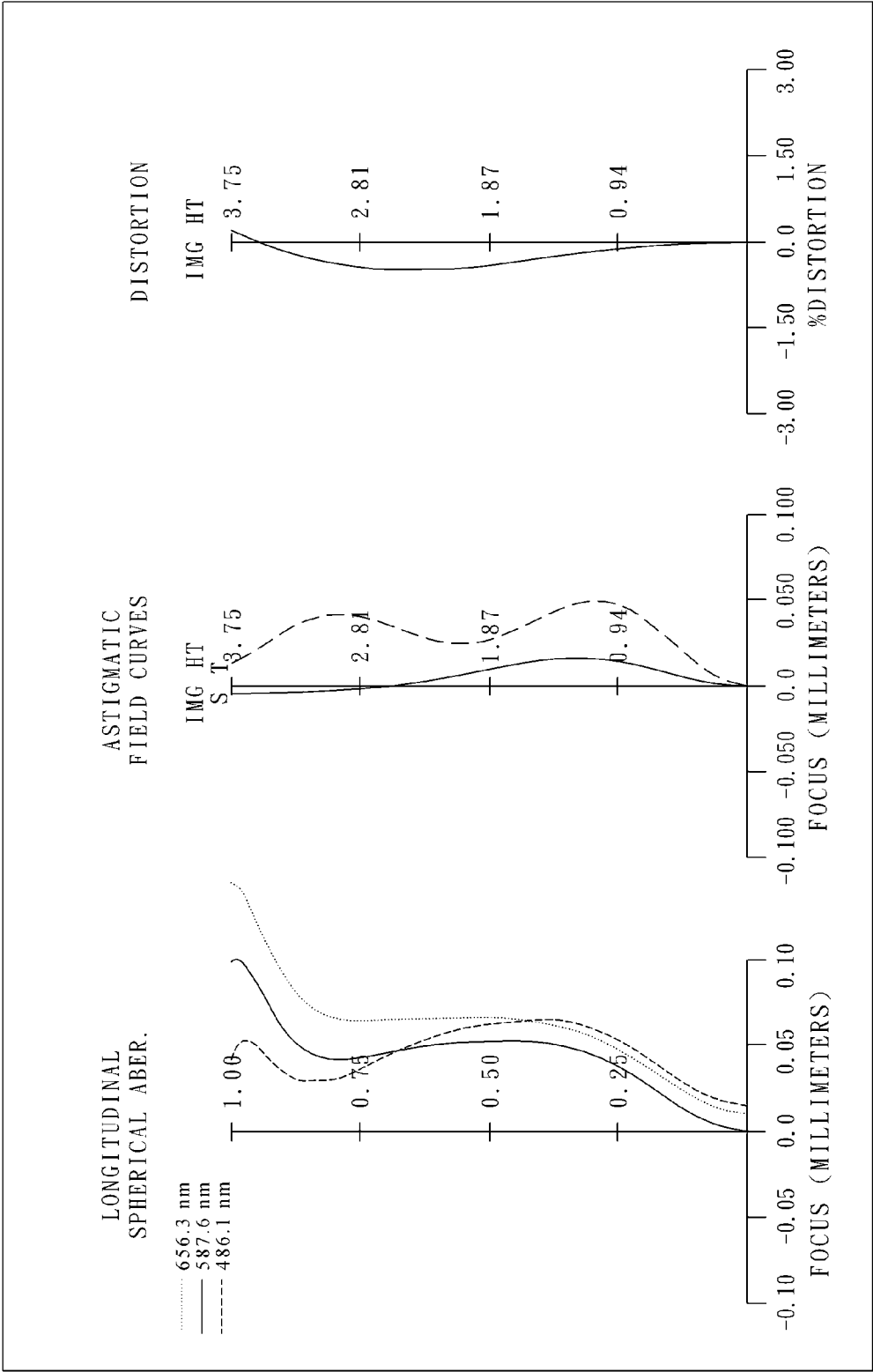


Fig. 3B

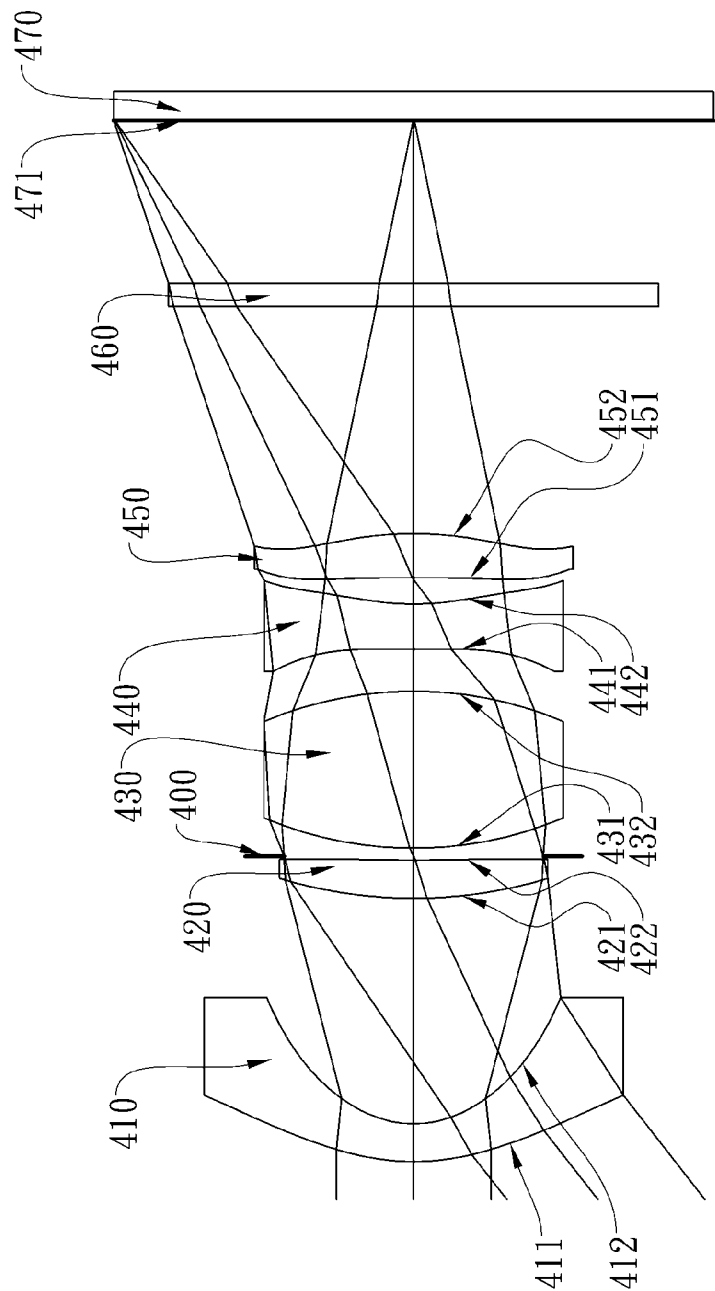


Fig. 4A

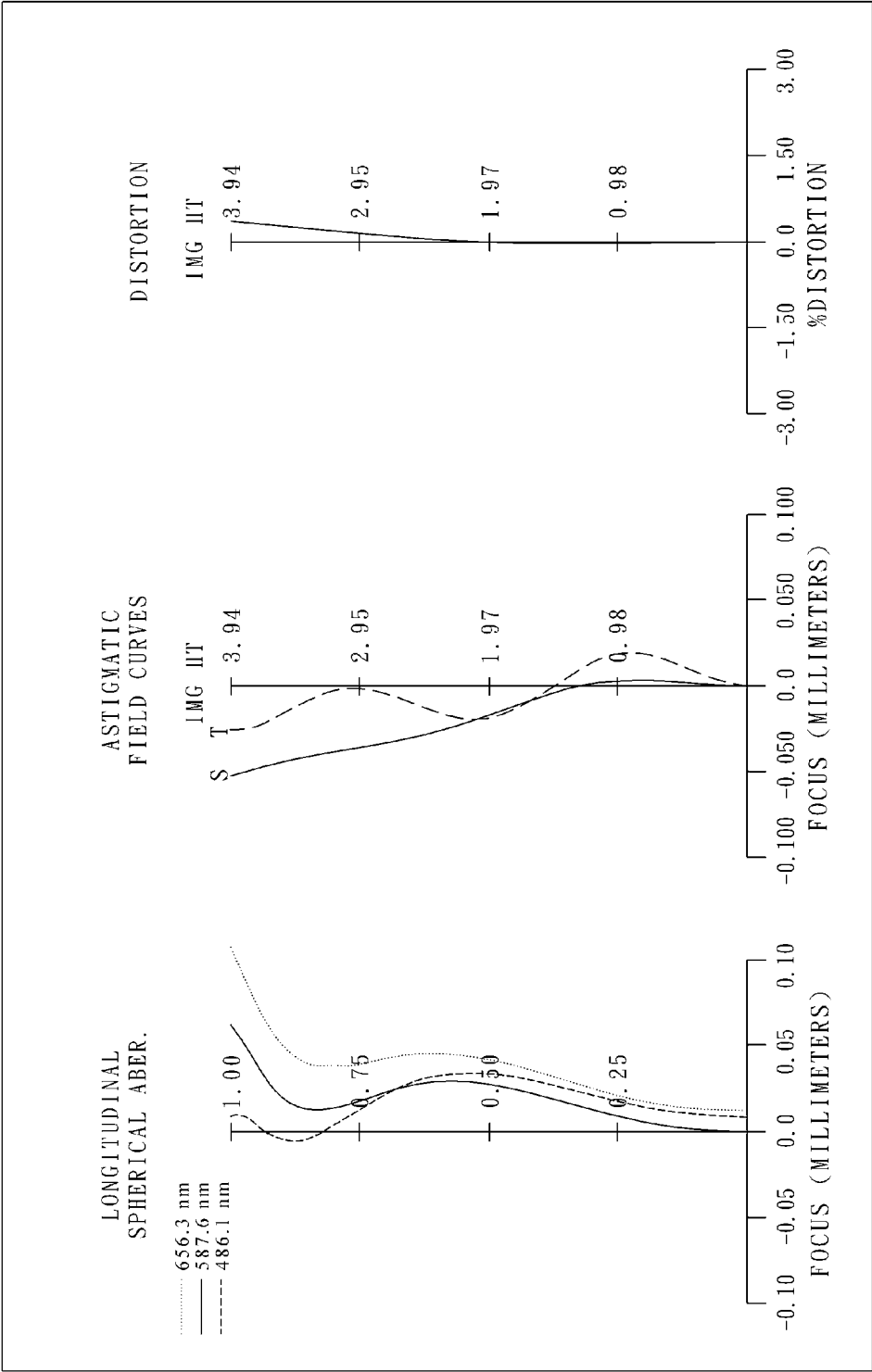


Fig. 4B

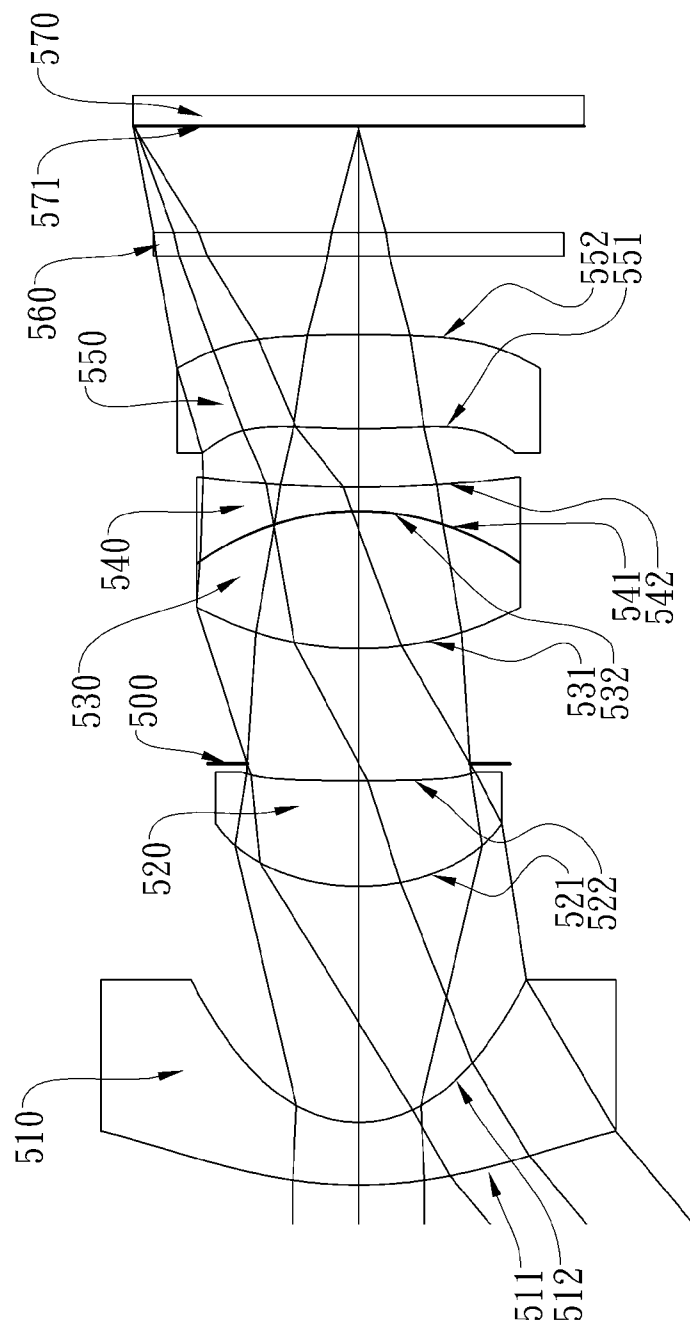


Fig. 5A

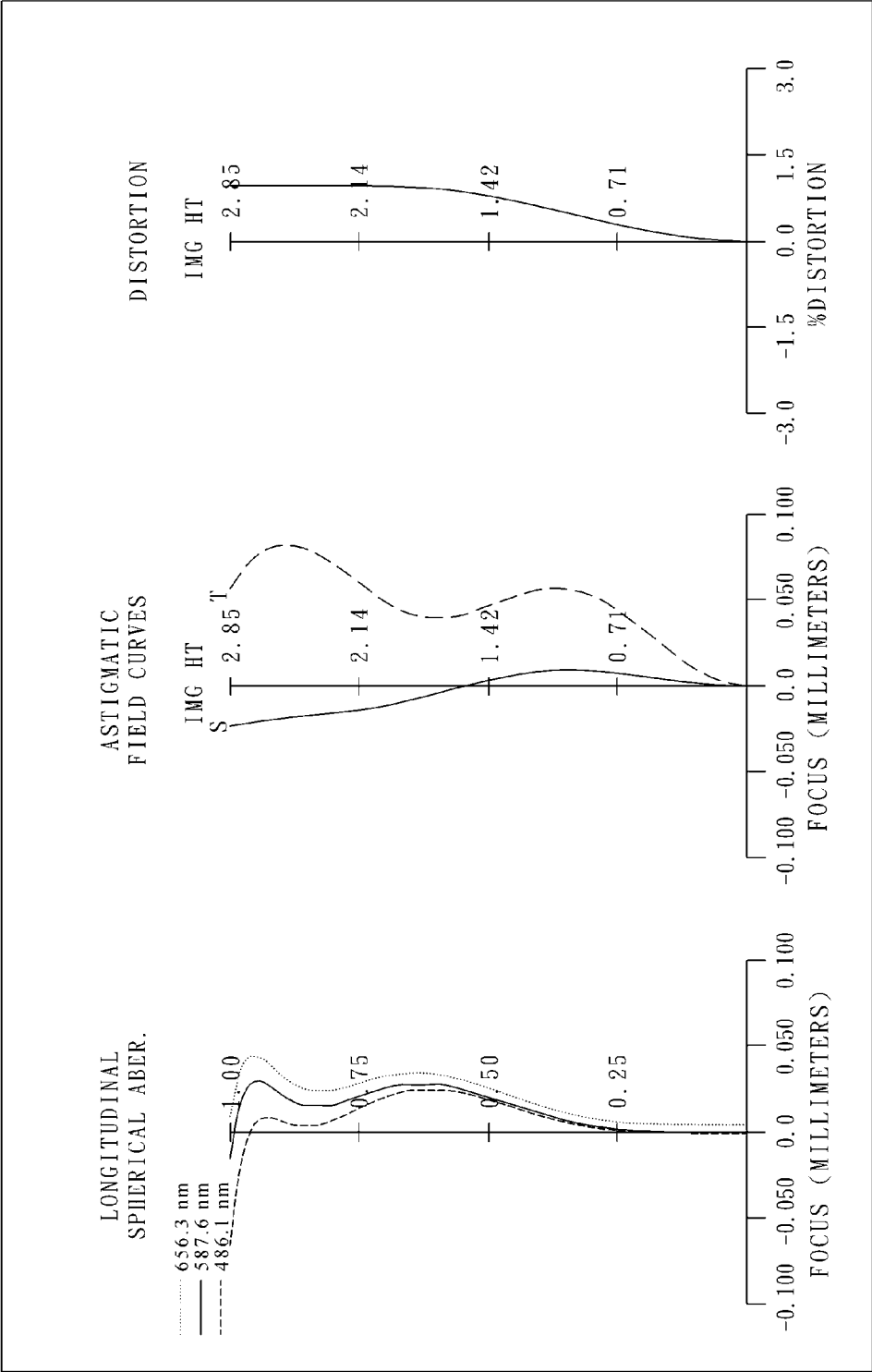


Fig. 5B

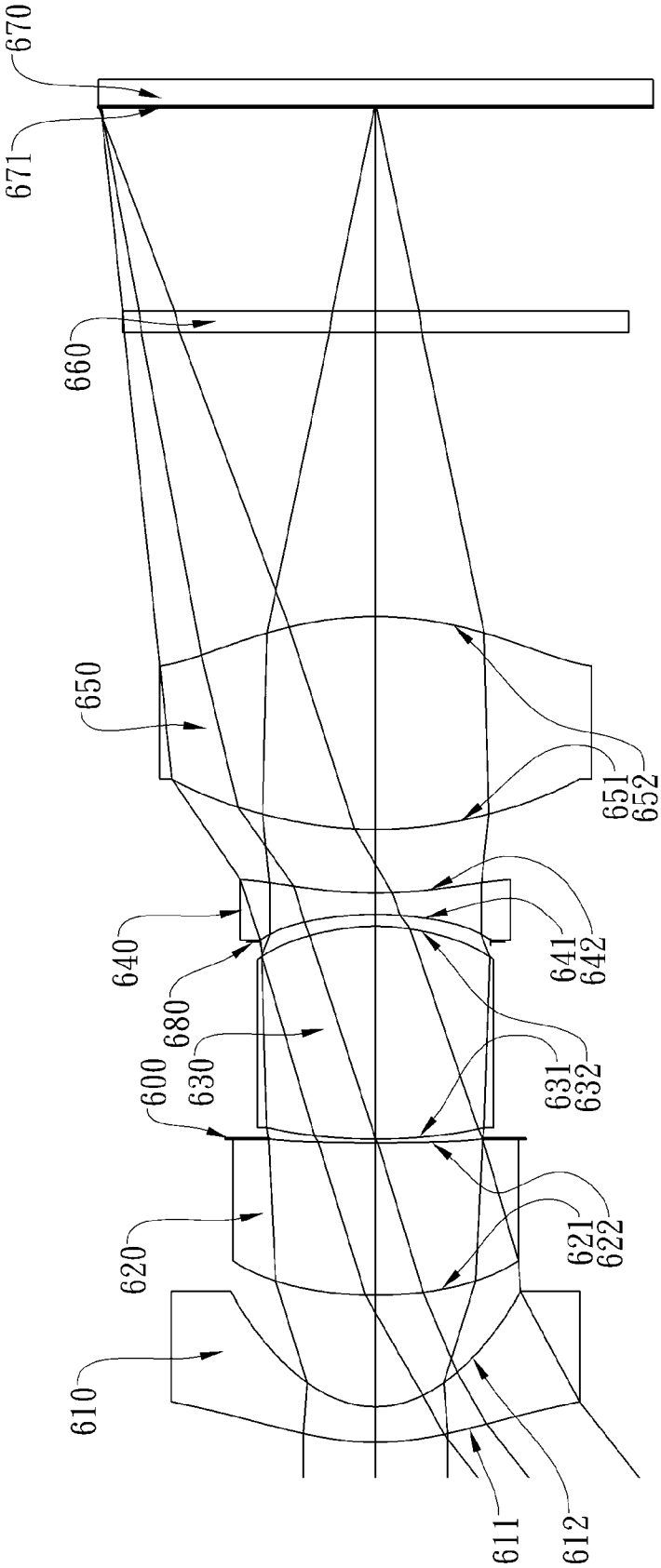


Fig. 6A

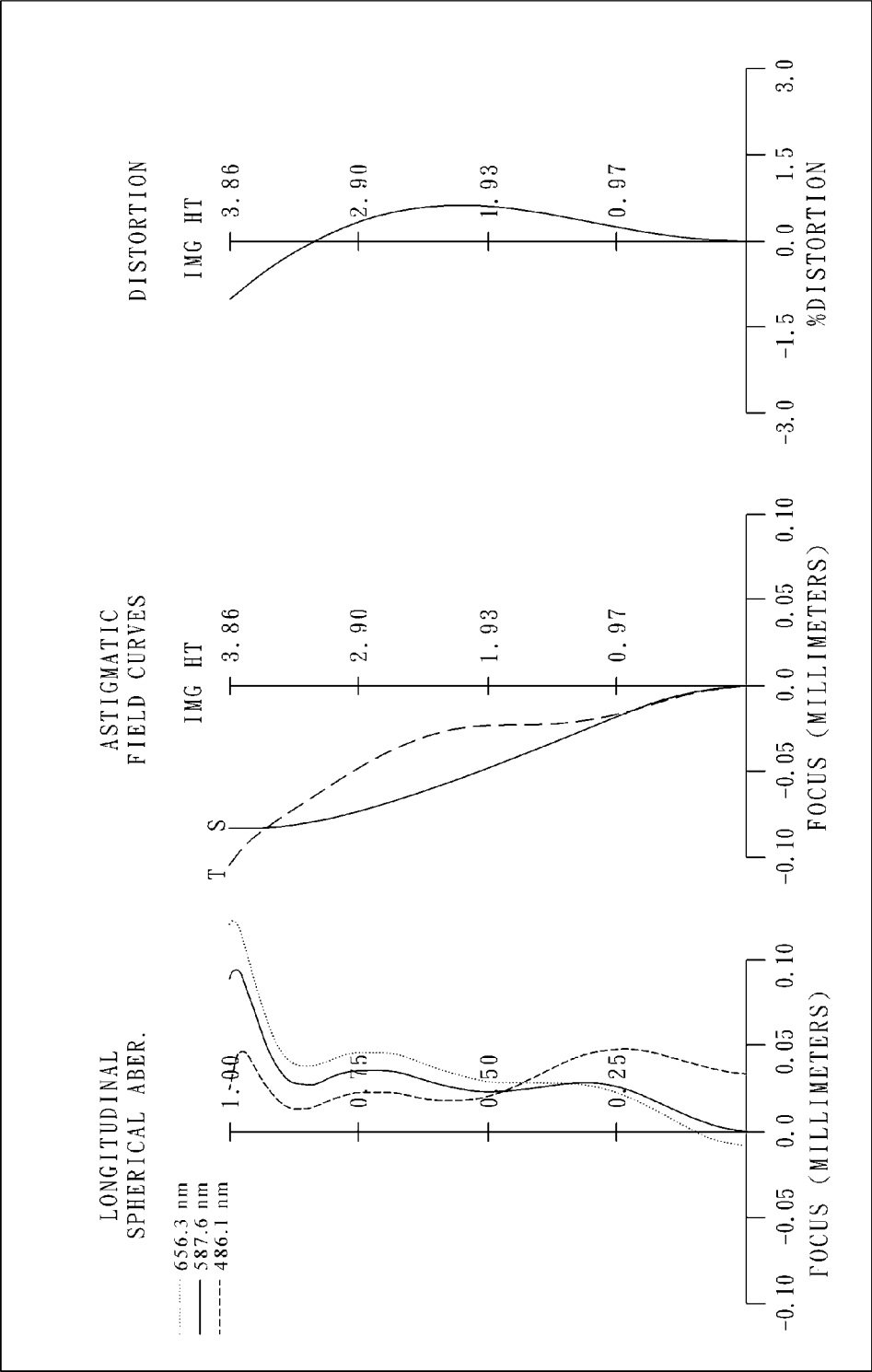


Fig. 6B

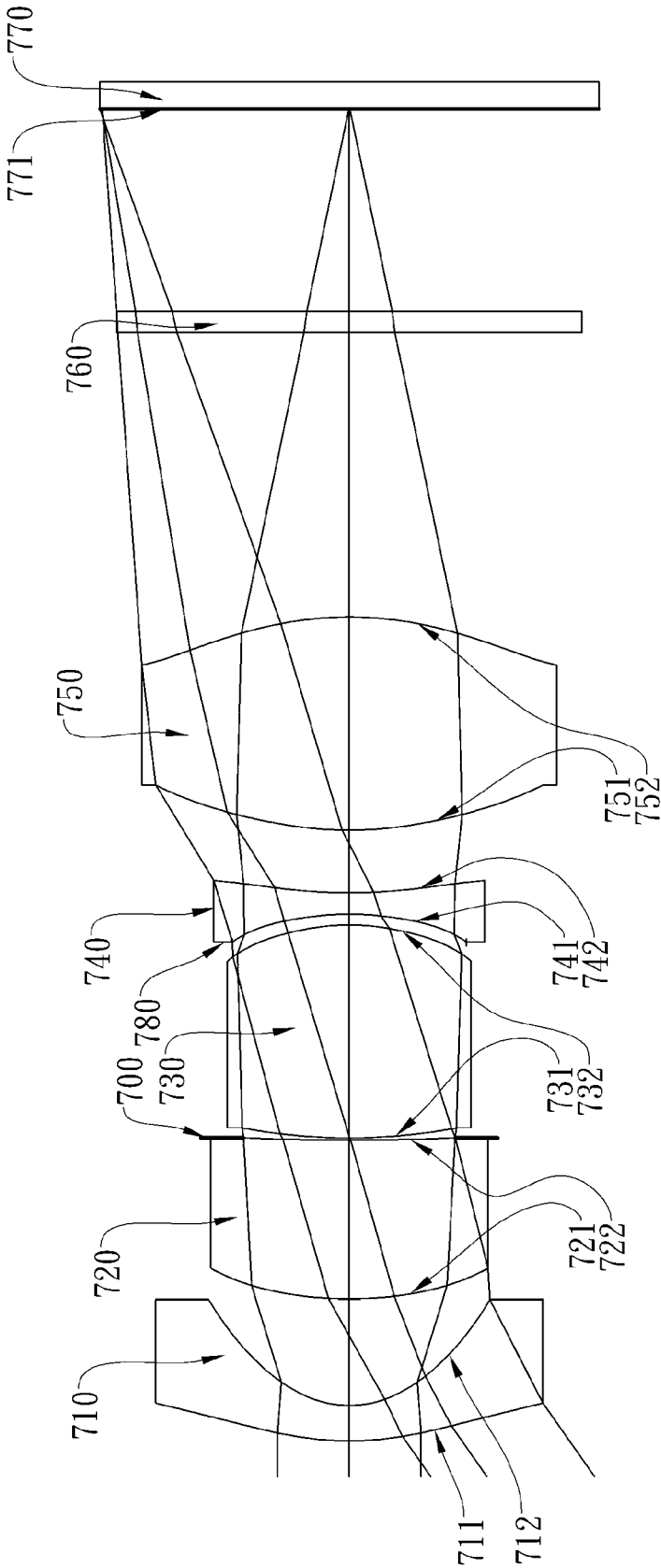


Fig. 7A

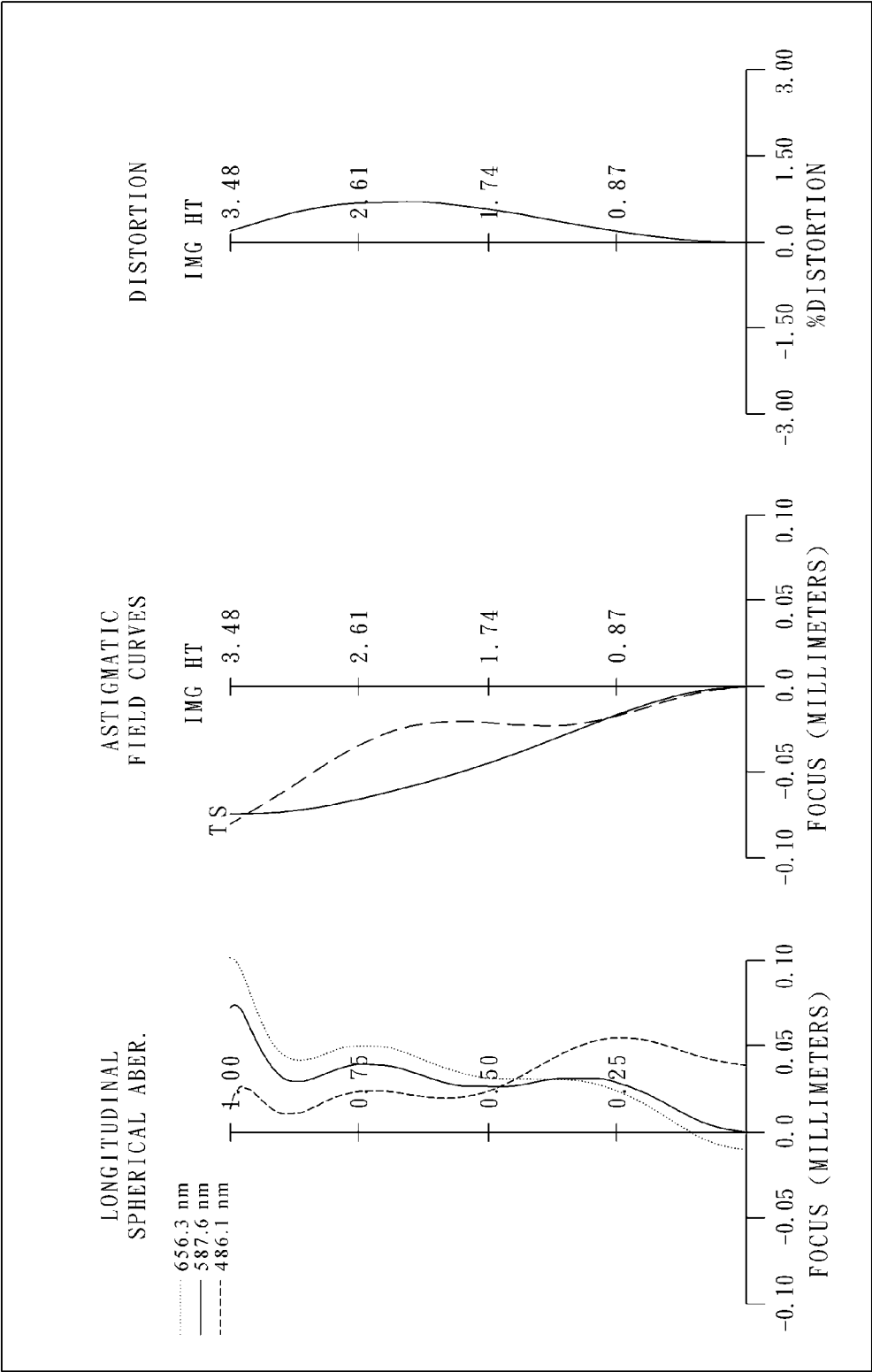


Fig. 7B

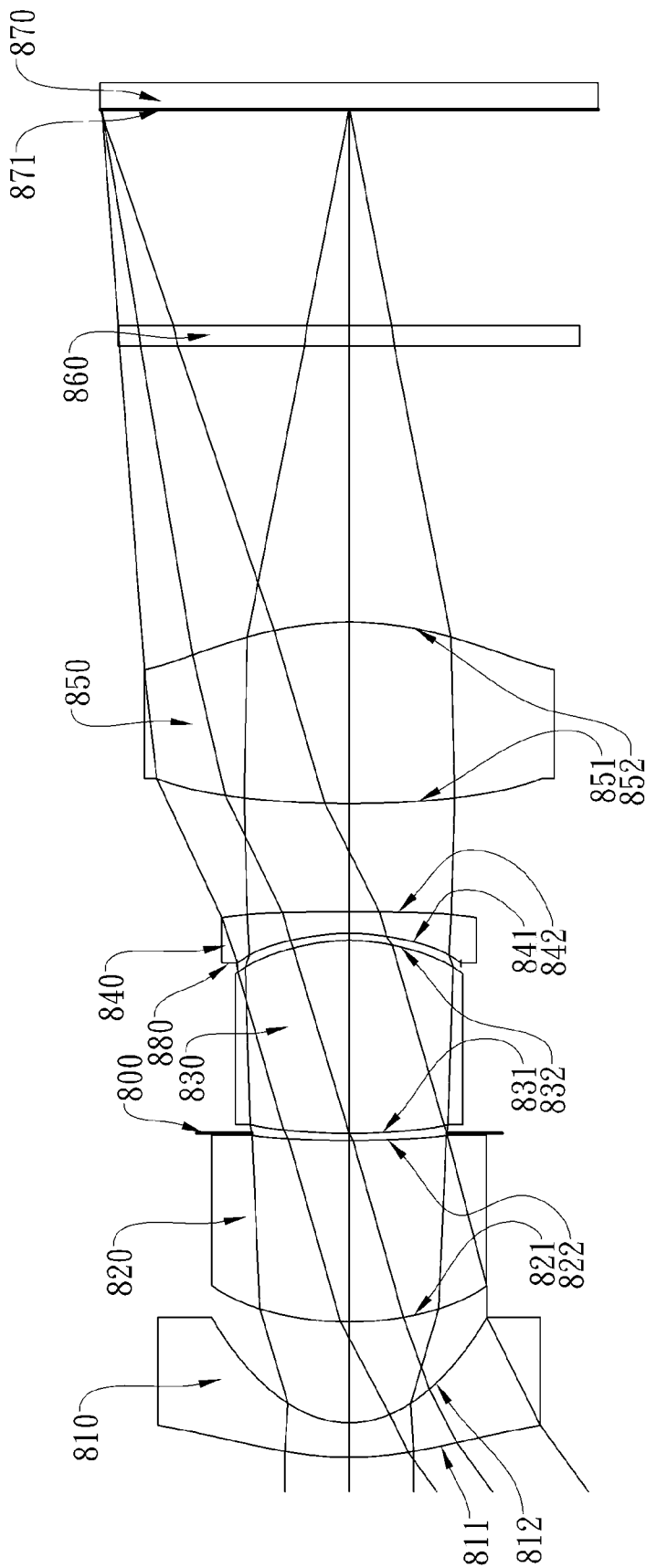


Fig. 8A

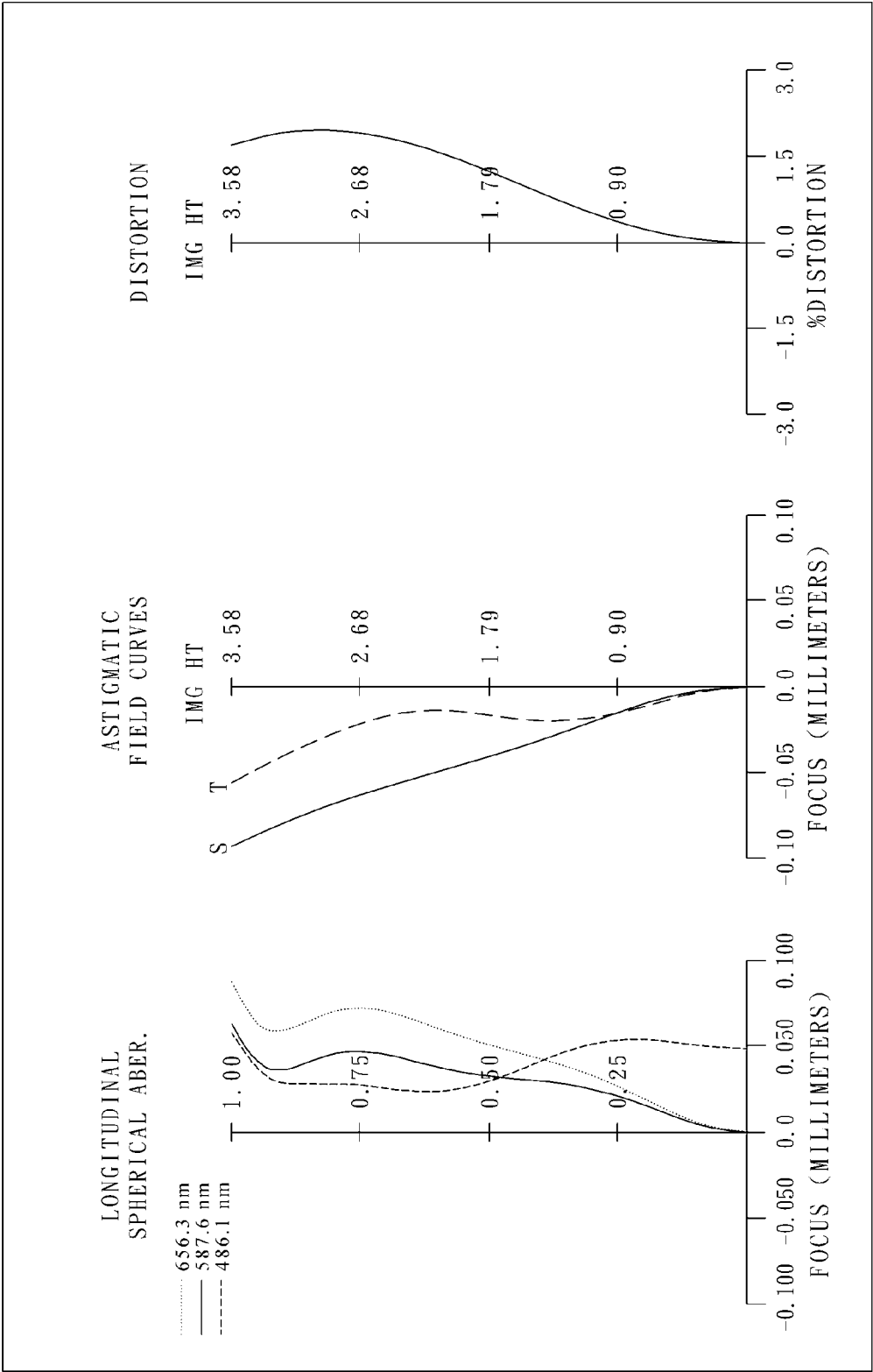


Fig. 8B

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IMAGE-CAPTURING OPTICAL LENS ASSEMBLY

CROSS-REFERENCE TO RELATED APPLICATIONS

This non-provisional application claims priority under 35 U.S.C. §119(a) on Patent Application No. 100119170 filed in Taiwan, R.O.C. on Jun. 1, 2011, the entire contents of which are hereby incorporated by reference.

BACKGROUND OF THE INVENTION

1. Field of the Invention

The present invention relates to an image-capturing optical lens assembly, and more particularly, to a compact image-capturing optical lens assembly used in an electronic product.

2. Description of the Prior Art

In recent years, optical imaging lenses have been used in a wider range of products, and in particular, they are used in the following products for example: mobile phone cameras, webcams, vehicle cameras, security surveillance cameras and electronic game devices. The sensor of a general imaging lens is none other than CCD (Charge Coupled Device) or CMOS Sensor (Complementary Metal Oxide Semiconductor Sensor). Furthermore, as advances in semiconductor manufacturing technology have allowed the pixel size of sensors to be reduced and imaging lenses have become more compact with high resolution, there is an increasing demand for imaging lenses featuring better image quality.

Imaging lenses used in mobile phone cameras, webcams, vehicle cameras, security surveillance cameras or electronic game devices typically require a larger field of view to capture an image of a larger area at one time. Generally, a conventional imaging lens assembly with a large angle of view, such as the four-element lens assembly disclosed in U.S. Pat. No. 7,446,955, is arranged in such manner that the front lens group has negative refractive power and the rear lens group has positive refractive power, thereby forming an inverse telephoto structure to achieve a wide field of view. While such arrangement facilitates the enlargement of the field of view, the aberration correction of the optical system is ineffective due to the inclusion of only one lens element in the rear lens group. Moreover, vehicles equipped with rear-view cameras have become more and more common, and there is a trend toward high-resolution, wide-angle lenses for rear-view cameras. Therefore, a need exists in the art for a wide-angle imaging lens assembly, which has a wide field of view, high image quality and a moderate total track length.

SUMMARY OF THE INVENTION

The present invention provides an image-capturing optical lens assembly comprising, in order from an object side to an image side, a front lens group, an aperture stop, and a rear lens group. The front lens group comprises, in order from the object side to the image side: a first lens element with negative refractive power having a convex object-side surface and a concave image-side surface; and a second lens element with positive refractive power having a convex object-side surface and a concave image-side surface. The rear lens group comprises, in order from the object side to the image side: a third lens element with positive refractive power having a convex object-side surface and a convex image-side surface; a fourth lens element with negative refractive power; and a fifth lens element with positive refractive power having a convex image-side surface, wherein a central thickness of the fifth

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lens element is CT5, a focal length of the lens assembly is f , a curvature radius of the object-side surface of the first lens element is $R1$, a curvature radius of the image-side surface of the fifth lens element is $R10$, and they satisfy the following relations: $0 < CT5/f < 0.7$; $0 < R1/f < 5.5$; and $R10/f < -0.85$.

With such an arrangement of optical elements, the optical system will have a field of view that is large enough; meanwhile, aberrations of the optical system can be favorably corrected to obtain good image quality.

In an image-capturing optical lens assembly of the present invention, the first lens element with negative refractive power allows the field of view of the system to be favorably enlarged. The second lens element and the third lens element with negative refractive power provide the main refractive power of the system, allowing the total track length of the lens assembly to be favorably reduced; also, the second lens element and the third lens element effectively distribute the refractive power of the system, thereby favorably reducing the sensitivity of the lens assembly. The fourth lens element with negative refractive power allows the aberration produced by the third lens element with positive refractive power to be corrected, and moreover, astigmatism of the system can be effectively corrected. The fifth lens element having positive refractive power acts as a correction lens that can correct the high-order aberration, and cooperates with the fourth lens element having negative refractive power to achieve telephoto effects; such an arrangement can help to shorten the back focal length and further to reduce the total track length of the lens assembly.

In the image-capturing optical lens assembly of the present invention, the first lens element with negative refractive power has a convex object-side surface and a concave image-side surface. With this configuration, the field of view of the system can be favorably enlarged, and the refraction of incident light is more moderate for preventing the aberration from being excessively large; therefore, it is favorable for obtaining a better balance between a wide field of view and aberration correction. As to the second lens element, it is a meniscus lens element having a convex object-side surface and a concave image-side surface; this configuration can help correct the astigmatism produced by the first lens element when the image-side surface of the first lens element is a concave surface with a higher curvature radius. The third lens element is a bi-convex lens element having convex object-side and image-side surfaces; with this configuration, the refractive power of the third lens element can be favorably distributed, thereby further reducing the total track length of the lens assembly and sensitivity of the system. The fifth lens element has a convex image-side surface; this configuration allows the angle at which light is projected onto the sensor from the off-axis field to be effectively reduced, thereby increasing the photosensitivity of the system.

In the image-capturing optical lens assembly of the present invention, the aperture stop is disposed between the front lens group and the rear lens group. In a wide-angle optical system, it is particularly needed to correct distortions and chromatic aberrations of magnification, and this can be done by placing the aperture stop at a position where the refractive power of the system is balanced. Therefore, the aperture stop is disposed between the front lens group and the rear lens group in the present invention. By doing so, the front lens group comprising at least two lens elements allows the system to have a sufficient field of view; meanwhile, the rear lens group comprising at least three lens elements allows aberrations of the system to be effectively reduced, thereby achieving a wide

field of view and good image quality at the same time. Also, such an arrangement of the aperture stop can help reduce the sensitivity of the system.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1A shows an image-capturing optical lens assembly in accordance with a first embodiment of the present invention.

FIG. 1B shows the aberration curves of the first embodiment of the present invention.

FIG. 2A shows an image-capturing optical lens assembly in accordance with a second embodiment of the present invention.

FIG. 2B shows the aberration curves of the second embodiment of the present invention.

FIG. 3A shows an image-capturing optical lens assembly in accordance with a third embodiment of the present invention.

FIG. 3B shows the aberration curves of the third embodiment of the present invention.

FIG. 4A shows an image-capturing optical lens assembly in accordance with a fourth embodiment of the present invention.

FIG. 4B shows the aberration curves of the fourth embodiment of the present invention.

FIG. 5A shows an image-capturing optical lens assembly in accordance with a fifth embodiment of the present invention.

FIG. 5B shows the aberration curves of the fifth embodiment of the present invention.

FIG. 6A shows an image-capturing optical lens assembly in accordance with a sixth embodiment of the present invention.

FIG. 6B shows the aberration curves of the sixth embodiment of the present invention.

FIG. 7A shows an image-capturing optical lens assembly in accordance with a seventh embodiment of the present invention.

FIG. 7B shows the aberration curves of the seventh embodiment of the present invention.

FIG. 8A shows an image-capturing optical lens assembly in accordance with an eighth embodiment of the present invention.

FIG. 8B shows the aberration curves of the eighth embodiment of the present invention.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

The present invention provides an image-capturing optical lens assembly comprising, in order from an object side to an image side, a front lens group, an aperture stop, and a rear lens group. The front lens group comprises, in order from the object side to the image side: a first lens element with negative refractive power having a convex object-side surface and a concave image-side surface; and a second lens element with positive refractive power having a convex object-side surface and a concave image-side surface. The rear lens group comprises, in order from the object side to the image side: a third lens element with positive refractive power having a convex object-side surface and a convex image-side surface; a fourth lens element with negative refractive power; and a fifth lens element with positive refractive power having a convex image-side surface, wherein a central thickness of the fifth lens element is CT5, a focal length of the lens assembly is f, a curvature radius of the object-side surface of the first lens

element is R1, a curvature radius of the image-side surface of the fifth lens element is R10, and they satisfy the following relations: $0 < CT5/f < 0.7$; $0 < R1/f < 5.5$; and $R10/f < -0.85$.

When the relation of $0 < CT5/f < 0.7$ is satisfied, the thickness of the fifth lens element is more favorable. As a result, a balance between improving the yield rate of the lens production process and correcting aberrations of the system can be favorably achieved; also, plastic lens products will be formed more easily with a higher homogeneity during the injection molding process.

When the relation of $0 < R1/f < 5.5$ is satisfied, the refraction of incident light is more moderate as the field of view of the system is enlarged, thereby preventing aberrations from increasing excessively.

When the relation of $R10/f < -0.85$ is satisfied, astigmatism of the system can be effectively corrected, thereby improving image quality of the system.

In the aforementioned image-capturing optical lens assembly, an axial distance between the first lens element and the second lens element is T12, an axial distance between the second lens element and the third lens element is T23, and preferably, they satisfy the following relation: $0 < T23/T12 < 0.65$. When this relation is satisfied, the axial distance between the second lens element and the third lens element and the distance between the first lens element and the second lens element are more favorable, so that the distances are not too short to make the assembling process difficult, or too long to affect minimizing the size of the lens assembly. More preferably, the following relation is satisfied: $0 < T23/T12 < 0.2$.

In the aforementioned image-capturing optical lens assembly, the focal length of the lens assembly is f, a focal length of the first lens element is f1, and preferably, they satisfy the following relation: $-1.6 < f/f1 < -0.6$. When this relation is satisfied, a balance between enlarging the field of view of the system and reducing the total track length of the lens assembly can be favorably achieved. More preferably, the following relation is satisfied: $-1.3 < f/f1 < -0.9$.

In the aforementioned image-capturing optical lens assembly, an axial distance between the aperture stop and an image plane is SL, an axial distance between the object-side surface of the first lens element and the image plane is TTL, and preferably, they satisfy the following relation: $0.55 < SL/TTL < 0.85$. When this relation is satisfied, the exit pupil of the lens assembly can be positioned far away from the image plane; thus, light will be projected onto the electronic sensor at a nearly perpendicular angle, and this is the telecentric feature of the image side. The telecentric feature is very important to the photosensitive ability of the current solid-state sensor because it can improve the photosensitivity of the sensor to reduce the probability of the occurrence of shading.

In the aforementioned image-capturing optical lens assembly, an Abbe number of the third lens element is V3, an Abbe number of the fourth lens element is V4, and preferably, they satisfy the following relation: $28 < V3 - V4 < 45$. When this relation is satisfied, the chromatic aberration of the lens assembly can be favorably corrected.

In the aforementioned image-capturing optical lens assembly, the focal length of the lens assembly is f, a focal length of the second lens element is f2, and preferably, they satisfy the following relation: $0.2 < f/f2 < 0.8$. When this relation is satisfied, the refractive power of the second lens element can be prevented from becoming too large; as a result, aberrations of the lens assembly can be favorably reduced and sensitivity of the system can be lowered. More preferably, the following relation is satisfied: $0.35 < f/f2 < 0.7$.

In the aforementioned image-capturing optical lens assembly, a curvature radius of the object-side surface of the second lens element is R3, a curvature radius of the image-side surface of the second lens element is R4, and preferably, they satisfy the following relation: $0 < R3/R4 < 0.5$. When this relation is satisfied, astigmatism of the system can be favorably corrected, and the refractive power can be properly adjusted; as a result, image resolution of the system can be increased, and aberrations of the lens assembly can be corrected.

In the aforementioned image-capturing optical lens assembly, the curvature radius of the object-side surface of the third lens element is R5, a curvature radius of the image-side surface of the third lens element is R6, and preferably, they satisfy the following relation: $-2.0 < R6/R5 < -0.2$. When this relation is satisfied, spherical aberrations of the system can be favorably corrected; as a result, the refractive power of the system can be properly distributed, thereby lowering the sensitivity of the system.

In the aforementioned image-capturing optical lens assembly, the focal length of the lens assembly is f, a focal length of the third lens element is f3, a focal length of the fourth lens element is f4, and preferably, they satisfy the following relation: $0 < f/f3 - |f/f4| < 0.5$. When this relation is satisfied, the refractive power of the third and fourth lens elements can be distributed more properly; as a result, the sensitivity of the system can be effectively controlled, and aberrations of the lens assembly can be corrected.

In the aforementioned image-capturing optical lens assembly, half of the maximum field of view of the lens assembly is HFOV, and preferably, it satisfies the following relation: $35 \text{ degrees} < \text{HFOV} < 45 \text{ degrees}$. When this relation is satisfied, the maximum field of view of the lens assembly is more favorable.

In the aforementioned image-capturing optical lens assembly, a central thickness of the third lens element is CT3, a central thickness of the fourth lens element is CT4, and preferably, they satisfy the following relation: $0 < CT4/CT3 < 0.3$. When this relation is satisfied, the thickness of either the third lens element or the fourth lens element is more favorable for assembling the lens assembly and arranging the space within it.

In the present image-capturing optical lens assembly, the lens elements can be made of glass or plastic material. If the lens elements are made of glass, the freedom for distributing the refractive power of the imaging optical lens assembly can be increased. If plastic material is adopted to produce the lens elements, the production cost will be reduced effectively. Moreover, the lens elements of the assembly can have aspheric surfaces. This allows more design parameters for the system since aspheric surfaces can be easily made into non-spherical profiles; also, it will reduce aberrations and the total number of lens elements, so that the total track length of the lens assembly can be effectively reduced.

In the present image-capturing optical lens assembly, if a lens element is described to have a convex surface, it means the portion of the surface in proximity to the optical axis is

convex; if a lens element is described to have a concave surface, it means the portion of the surface in proximity to the optical axis is concave.

In the present image-capturing optical lens assembly, there can be at least one aperture stop, such as a glare stop or field stop, provided for eliminating stray light, thereby promoting image resolution of the lens assembly.

Preferred embodiments of the present invention will be described in the following paragraphs by referring to the accompanying drawings.

Embodiment 1

FIG. 1A shows an image-capturing optical lens assembly in accordance with the first embodiment of the present invention, and FIG. 1B shows the aberration curves of the first embodiment of the present invention. The image-capturing optical lens assembly in the first embodiment mainly comprises five lens elements, in order from an object side to an image side: a front lens group, an aperture stop 100 and a rear lens group, wherein:

the front lens group comprises, in order from the object side to the image side:

a plastic first lens element 110 with negative refractive power having a convex object-side surface 111 and a concave image-side surface 112, both of the surfaces 111 and 112 being aspheric; and

a plastic second lens element 120 with positive refractive power having a convex object-side surface 121 and a concave image-side surface 122, both of the surfaces 121 and 122 being aspheric;

the rear lens group comprises, in order from the object side to the image side:

a glass third lens element 130 with positive refractive power having a convex object-side surface 131 and a convex image-side surface 132;

a plastic fourth lens element 140 with negative refractive power having a convex object-side surface 141 and a concave image-side surface 142, both of the surfaces 141 and 142 being aspheric; and

a plastic fifth lens element 150 with positive refractive power having a convex object-side surface 151 and a convex image-side surface 152, both of the surfaces 151 and 152 being aspheric.

Moreover, the image-capturing optical lens assembly comprises an IR filter 160 disposed between the image-side surface 152 of the fifth lens element 150 and an image plane 171; the IR filter 160 is made of glass and has no influence on the focal length of the image-capturing optical lens assembly. The image-capturing optical lens assembly further comprises an image sensor 170 provided on the image plane 171.

The detailed optical data of the first embodiment is shown in TABLE 1, and the aspheric surface data is shown in TABLE 2, wherein the units of the curvature radius, the thickness and the focal length are expressed in mm, and HFOV is half of the maximum field of view.

TABLE 1

(Embodiment 1)							
f = 5.05 mm, Fno = 2.58, HFOV = 36.1 deg.							
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal length
0	Object	Plano	Infinity				
1	Lens 1	4.958200 (ASP)	0.500	Plastic	1.544	55.9	-4.37

TABLE 1-continued

(Embodiment 1)							
f = 5.05 mm, Fno = 2.58, HFOV = 36.1 deg.							
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal length
2		1.550560 (ASP)	2.190				
3	Lens 2	4.863100 (ASP)	1.417	Plastic	1.544	55.9	9.39
4		90.909100 (ASP)	0.150				
5	Ape. Stop	Plano	-0.050				
6	Lens 3	5.636	2.999	Glass	1.620	60.3	5.06
7		-5.636	0.187				
8	Lens 4	11.216700 (ASP)	0.318	Plastic	1.634	23.8	-5.78
9		2.731180 (ASP)	0.210				
10	Lens 5	9.114500 (ASP)	0.711	Plastic	1.544	55.9	7.28
11		-6.818800 (ASP)	3.000				
12	IR-filter	Plano	0.300	Glass	1.516	64.1	—
13		Plano	2.859				
14	Image	Plano	—				

Reference wavelength is 587.6 nm (d-line)

TABLE 2

Aspheric Coefficients				
Surface #	1	2	3	4
k =	-2.66212E+00	-9.85189E-01	0.00000E+00	1.81788E+02
A4 =	-9.91825E-04	1.93492E-02	8.02061E-03	4.70355E-03
A6 =	-1.06236E-03	-1.05880E-03	1.70387E-03	2.56067E-03
A8 =	1.50007E-04	1.93798E-04	-4.65386E-04	-8.41304E-04
A10 =	-7.33746E-06	-3.18585E-05	1.01368E-04	2.53861E-04
Surface #	8	9	10	11
k =	-9.90000E+01	-7.57922E+00	-1.00000E+00	-6.57179E+01
A4 =	-3.57307E-02	-2.34316E-03	1.31669E-02	-1.04651E-02
A6 =	9.47609E-03	1.71494E-03	3.62072E-04	9.49392E-03
A8 =	-9.33684E-04	3.60013E-04	-2.27327E-04	-1.86708E-03
A10 =	-5.84118E-05	-1.10415E-04	1.39560E-05	1.55424E-04
A12 =	1.10023E-05			
A14 =	-4.17134E-07			

The equation of the aspheric surface profiles is expressed as follows:

$$X(Y) = (Y^2/R)/(1 + \sqrt{1 - (1 + k)(Y/R)^2}) + \sum_i (Ai)^*(Y^i)$$

wherein:

X: the height of a point on the aspheric surface at a distance Y from the optical axis relative to the tangential plane at the aspheric surface vertex;

Y: the distance from the point on the curve of the aspheric surface to the optical axis;

k: the conic coefficient;

Ai: the aspheric coefficient of order i.

In the first embodiment of the present image-capturing optical lens assembly, the focal length of the lens assembly is f, and it satisfies the relation: f=5.05 (mm).

In the first embodiment of the present image-capturing optical lens assembly, the f-number of the lens assembly is Fno, and it satisfies the relation: Fno=2.58.

In the first embodiment of the present image-capturing optical lens assembly, half of the maximum field of view of the lens assembly is HFOV, and it satisfies the relation: HFOV=36.1 (degrees).

In the first embodiment of the present image-capturing optical lens assembly, the Abbe number of the third lens

element **130** is V3, the Abbe number of the fourth lens element **140** is V4, and they satisfy the relation: V3-V4=36.5.

In the first embodiment of the present image-capturing optical lens assembly, the central thickness of the fifth lens element **150** is CT5, the focal length of the lens assembly is f, and they satisfy the relation: CT5/f=0.14.

In the first embodiment of the present image-capturing optical lens assembly, the central thickness of the third lens element **130** is CT3, the central thickness of the fourth lens element **140** is CT4, and they satisfy the relation: CT4/CT3=0.11.

In the first embodiment of the present image-capturing optical lens assembly, the axial distance between the first lens element **110** and the second lens element **120** is T12, the axial distance between the second lens element **120** and the third lens element **130** is T23, and they satisfy the relation: T23/T12=0.05.

In the first embodiment of the present image-capturing optical lens assembly, the focal length of the lens assembly is f, the curvature radius of the object-side surface **111** of the fifth lens element **110** is R1, and they satisfy the relation: R1/f=0.98.

In the first embodiment of the present image-capturing optical lens assembly, the focal length of the lens assembly is f, the curvature radius of the image-side surface **152** of the fifth lens element **150** is R10, and they satisfy the relation: R10/f=-1.35.

In the first embodiment of the present image-capturing optical lens assembly, the curvature radius of the object-side

surface **121** of the second lens element **120** is R3, the curvature radius of the image-side surface **122** of the second lens element **120** is R4, and they satisfy the relation: $R3/R4=0.05$.

In the first embodiment of the present image-capturing optical lens assembly, the curvature radius of the object-side surface **131** of the third lens element **130** is R5, the curvature radius of the image-side surface **132** of the third lens element **130** is R6, and they satisfy the relation: $R6/R5=-1.00$.

In the first embodiment of the present image-capturing optical lens assembly, the focal length of the lens assembly is f , the focal length of the first lens element **110** is $f1$, and they satisfy the relation: $f/f1=-1.16$.

In the first embodiment of the present image-capturing optical lens assembly, the focal length of the lens assembly is f , the focal length of the second lens element **120** is $f2$, and they satisfy the relation: $f/f2=0.54$.

In the first embodiment of the present image-capturing optical lens assembly, the focal length of the lens assembly is f , the focal length of the third lens element **130** is $f3$, the focal length of the fourth lens element **140** is $f4$, and they satisfy the relation: $f/f3-|f/f4|=0.13$.

In the first embodiment of the present image-capturing optical lens assembly, the axial distance between the aperture stop **100** and the image plane **171** is SL, the axial distance between the object-side surface **111** of the first lens element **110** and the image plane **171** is TTL, and they satisfy the following relation: $SL/TTL=0.71$.

Embodiment 2

FIG. 2A shows an image-capturing optical lens assembly in accordance with the second embodiment of the present invention, and FIG. 2B shows the aberration curves of the second embodiment of the present invention. The image-capturing optical lens assembly in the second embodiment mainly comprises five lens elements, in order from an object

side to an image side: a front lens group, an aperture stop **200** and a rear lens group, wherein:

the front lens group comprises, in order from the object side to the image side:

a plastic first lens element **210** with negative refractive power having a convex object-side surface **211** and a concave image-side surface **212**, both of the surfaces **211** and **212** being aspheric; and

a plastic second lens element **220** with positive refractive power having a convex object-side surface **221** and a concave image-side surface **222**, both of the surfaces **221** and **222** being aspheric;

the rear lens group comprises, in order from the object side to the image side:

a glass third lens element **230** with positive refractive power having a convex object-side surface **231** and a convex image-side surface **232**;

a plastic fourth lens element **240** with negative refractive power having a concave object-side surface **241** and a concave image-side surface **242**, both of the surfaces **241** and **242** being aspheric; and

a plastic fifth lens element **250** with positive refractive power having a convex object-side surface **251** and a convex image-side surface **252**, both of the surfaces **251** and **252** being aspheric.

Moreover, the image-capturing optical lens assembly comprises an IR filter **260** disposed between the image-side surface **252** of the fifth lens element **250** and an image plane **271**; the IR filter **260** is made of glass and has no influence on the focal length of the image-capturing optical lens assembly. The image-capturing optical lens assembly further comprises an image sensor **270** provided on the image plane **271**.

The detailed optical data of the second embodiment is shown in TABLE 3, and the aspheric surface data is shown in TABLE 4, wherein the units of the curvature radius, the thickness and the focal length are expressed in mm, and HFOV is half of the maximum field of view.

TABLE 3

(Embodiment 2)							
$f = 5.06$ mm, $Fno = 2.60$, $HFOV = 37.6$ deg.							
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal length
0	Object	Plano	Infinity				
1	Lens 1	4.900400 (ASP)	0.500	Plastic	1.544	55.9	-4.62
2		1.602890 (ASP)	2.584				
3	Lens 2	4.474300 (ASP)	0.719	Plastic	1.544	55.9	9.74
4		27.027000 (ASP)	0.063				
5	Ape. Stop	Plano	0.000				
6	Lens 3	5.881	2.861	Glass	1.620	60.3	5.23
7		-5.881	0.249				
8	Lens 4	-32.258100 (ASP)	0.400	Plastic	1.634	23.8	-4.77
9		3.354500 (ASP)	0.220				
10	Lens 5	5.011800 (ASP)	1.021	Plastic	1.544	55.9	5.85
11		-8.091100 (ASP)	3.000				
12	IR-filter	Plano	0.300	Glass	1.516	64.1	—
13		Plano	2.750				
14	Image	Plano	—				

Reference wavelength is 587.6 nm (d-line)

TABLE 4

Aspheric Coefficients				
Surface #	1	2	3	4
k =	-9.19767E-01	-9.14913E-01	0.00000E+00	-9.90000E+01
A4 =	7.93210E-04	1.97695E-02	1.78951E-03	-2.58009E-03

TABLE 4-continued

Aspheric Coefficients				
A6 =	-1.11658E-03	-1.01271E-03	1.84376E-03	2.65280E-03
A8 =	1.38186E-04	4.77678E-04	-9.11364E-04	-1.34106E-03
A10 =	-6.70122E-06	-3.06769E-05	1.83308E-04	2.63586E-04
Surface #	8	9	10	11
k =	9.90000E+01	-1.05548E+01	-8.78235E+00	-9.90000E+01
A4 =	-3.78139E-02	-8.05331E-03	8.56318E-03	-9.80453E-03
A6 =	7.89218E-03	1.25618E-03	2.06395E-04	8.43274E-03
A8 =	-1.07829E-03	3.59354E-04	-1.76958E-04	-1.67425E-03
A10 =	-4.02626E-06	-8.84571E-05	2.91877E-05	1.41778E-04
A12 =	3.89571E-06			
A14 =	-3.16608E-06			

The equation of the aspheric surface profiles of the second embodiment has the same form as that of the first embodiment. Moreover, the definitions of the parameters of the relations are the same as those set forth in the first embodiment, but the value of the relations of the second embodiment are listed in the following TABLE 5:

TABLE 5

(Embodiment 2)

f	5.06
Fno	2.60
HFOV	37.6
V3-V4	36.5
CT5/f	0.20
CT4/CT3	0.14
T23/T12	0.02
R1/f	0.97
R10/f	-1.60
R3/R4	0.17
R6/R5	-1.00
f/f1	-1.10
f/f2	0.52
f/f3- f/f4	-0.09
SL/TTL	0.74

Embodiment 3

FIG. 3A shows an image-capturing optical lens assembly in accordance with the third embodiment of the present invention, and FIG. 3B shows the aberration curves of the third embodiment of the present invention. The image-capturing optical lens assembly in the third embodiment mainly comprises five lens elements, in order from an object side to an image side: a front lens group, an aperture stop **300** and a rear lens group, wherein:

the front lens group comprises, in order from the object side to the image side:

a plastic first lens element **310** with negative refractive power having a convex object-side surface **311** and a concave image-side surface **312**, both of the surfaces **311** and **312** being aspheric; and

a plastic second lens element **320** with positive refractive power having a convex object-side surface **321** and a concave image-side surface **322**, both of the surfaces **321** and **322** being aspheric;

the rear lens group comprises, in order from the object side to the image side:

a glass third lens element **330** with positive refractive power having a convex object-side surface **331** and a convex image-side surface **332**;

a plastic fourth lens element **340** with negative refractive power having a convex object-side surface **341** and a concave image-side surface **342**, both of the surfaces **341** and **342** being aspheric; and

a plastic fifth lens element **350** with positive refractive power having a convex object-side surface **351** and a convex image-side surface **352**, both of the surfaces **351** and **352** being aspheric.

Moreover, the image-capturing optical lens assembly comprises an IR filter **360** disposed between the image-side surface **352** of the fifth lens element **350** and an image plane **371**; the IR filter **360** is made of glass and has no influence on the focal length of the image-capturing optical lens assembly. The image-capturing optical lens assembly further comprises an image sensor **370** provided on the image plane **371**.

The detailed optical data of the third embodiment is shown in TABLE 6, and the aspheric surface data is shown in TABLE 7, wherein the units of the curvature radius, the thickness and the focal length are expressed in mm, and HFOV is half of the maximum field of view.

TABLE 6

(Embodiment 3)							
f = 4.97 mm, Fno = 2.45, HFOV = 36.8 deg.							
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal length
0	Object	Plano	Infinity				
1	Lens 1	3.903100 (ASP)	0.500	Plastic	1.544	55.9	-4.98
2		1.527190 (ASP)	2.625				
3	Lens 2	5.843600 (ASP)	0.731	Plastic	1.544	55.9	12.51
4		39.525700 (ASP)	0.087				
5	Ape. Stop	Plano	0.265				
6	Lens 3	5.106	1.985	Glass	1.620	60.3	4.45
7		-5.106	0.558				
8	Lens 4	6.117100 (ASP)	0.379	Plastic	1.650	21.4	-5.74

TABLE 6-continued

(Embodiment 3)							
$f = 4.97$ mm, $Fno = 2.45$, $HFOV = 36.8$ deg.							
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal length
9		2.260930 (ASP)	0.273				
10	Lens 5	15.518200 (ASP)	0.761	Plastic	1.544	55.9	8.83
11		-6.840000 (ASP)	3.000				
12	IR-filter	Plano	0.300	Glass	1.516	64.1	—
13		Plano	2.275				
14	Image	Plano	—				

Reference wavelength is 587.6 nm (d-line)

TABLE 7

Aspheric Coefficients				
Surface #	1	2	3	4
k =	-9.67365E+00	-1.68564E+00	1.67011E-01	9.90000E+01
A4 =	2.98561E-03	2.38590E-02	-4.08657E-03	-7.00761E-03
A6 =	1.04816E-04	3.86679E-03	2.45400E-03	2.14683E-03
A8 =	-3.37644E-05	-3.41349E-04	-7.78605E-04	-7.24526E-04
A10 =	1.10343E-06	1.58511E-04	1.68737E-04	1.84945E-04
Surface #	8	9	10	11
k =	-9.41194E+01	-9.75135E+00	-1.00000E+00	-5.04290E+01
A4 =	-4.55826E-02	-2.01345E-02	-1.58320E-02	-1.32994E-02
A6 =	7.60709E-03	2.30888E-03	5.80290E-03	7.71855E-03
A8 =	-8.04223E-04	9.26813E-04	5.16494E-04	-5.14221E-04
A10 =	2.92612E-05	-9.83143E-05	-1.39156E-04	-1.72975E-05
A12 =	1.03020E-05			
A14 =	-4.43366E-07			

The equation of the aspheric surface profiles of the third embodiment has the same form as that of the first embodiment. Moreover, the definitions of the parameters of the relations are the same as those set forth in the first embodiment, but the value of the relations of the third embodiment are listed in the following TABLE 8:

TABLE 8

(Embodiment 3)	
f	4.97
Fno	2.45
HFOV	36.8
V3-V4	38.9
CT5/f	0.15
CT4/CT3	0.19
T23/T12	0.13
R1/f	0.79
R10/f	-1.38
R3/R4	0.15
R6/R5	-1.00
f/f1	-1.00
f/f2	0.40
f/f3- f/f4	0.25
SL/TTL	0.71

Embodiment 4

FIG. 4A shows an image-capturing optical lens assembly in accordance with the fourth embodiment of the present invention, and FIG. 4B shows the aberration curves of the fourth embodiment of the present invention. The image-capturing optical lens assembly in the fourth embodiment mainly comprises five lens elements, in order from an object side to an image side: a front lens group, an aperture stop **400** and a rear lens group, wherein:

the front lens group comprises, in order from the object side to the image side:

a plastic first lens element **410** with negative refractive power having a convex object-side surface **411** and a concave image-side surface **412**, both of the surfaces **411** and **412** being aspheric; and

a plastic second lens element **420** with positive refractive power having a convex object-side surface **421** and a concave image-side surface **422**, both of the surfaces **421** and **422** being aspheric;

the rear lens group comprises, in order from the object side to the image side:

a glass third lens element **430** with positive refractive power having a convex object-side surface **431** and a convex image-side surface **432**;

a plastic fourth lens element **440** with negative refractive power having a convex object-side surface **441** and a concave image-side surface **442**, both of the surfaces **441** and **442** being aspheric; and

a plastic fifth lens element **450** with positive refractive power having a concave object-side surface **451** and a convex image-side surface **452**, both of the surfaces **451** and **452** being aspheric.

Moreover, the image-capturing optical lens assembly comprises an IR filter **460** disposed between the image-side surface **452** of the fifth lens element **450** and an image plane **471**; the IR filter **460** is made of glass and has no influence on the focal length of the image-capturing optical lens assembly. The image-capturing optical lens assembly further comprises an image sensor **470** provided on the image plane **471**.

The detailed optical data of the fourth embodiment is shown in TABLE 9, and the aspheric surface data is shown in TABLE 10, wherein the units of the curvature radius, the thickness and the focal length are expressed in mm, and HFOV is half of the maximum field of view.

TABLE 9

(Embodiment 4)								
f = 5.00 mm, Fno = 2.45, HFOV = 38.1 deg.								
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal length	
0	Object	Plano	Infinity					
1	Lens 1	2.941270 (ASP)	0.500	Plastic	1.544	55.9	-5.32	
2		1.371400 (ASP)	2.976					
3	Lens 2	5.730700 (ASP)	0.502	Plastic	1.544	55.9	13.74	
4		23.809500 (ASP)	0.050					
5	Ape. Stop	Plano	0.109					
6	Lens 3	5.079	2.073	Glass	1.620	60.3	4.44	
7		-5.079	0.550					
8	Lens 4	20.000000 (ASP)	0.600	Plastic	1.650	21.4	-5.93	
9		3.191700 (ASP)	0.347					
10	Lens 5	-38.461500 (ASP)	0.583	Plastic	1.544	55.9	9.38	
11		-4.529900 (ASP)	3.000					
12	IR-filter	Plano	0.300	Glass	1.516	64.1	—	
13		Plano	2.151					
14	Image	Plano	—					

Reference wavelength is 587.6 nm (d-line)

TABLE 10

Aspheric Coefficients				
Surface #	1	2	3	4
k =	-4.71822E+00	-1.55203E+00	-6.72801E-01	-8.50842E+01
A4 =	1.50564E-03	2.65024E-02	-4.66217E-03	-8.89413E-03
A6 =	1.46424E-04	2.29291E-03	2.53702E-03	2.32350E-03
A8 =	-5.95934E-05	1.35079E-04	-9.49677E-04	-8.85654E-04
A10 =	3.02944E-06	3.55351E-05	1.66026E-04	1.72681E-04
Surface #	8	9	10	11
k =	-8.72604E+01	-1.03828E+01	+1.00000E+00	-1.88095E+01
A4 =	-4.82135E-02	-1.89238E-02	-8.82586E-03	-1.94206E-02
A6 =	6.84826E-03	1.48434E-03	4.69088E-03	9.14502E-03
A8 =	-9.04923E-04	7.48503E-04	4.87148E-04	-3.55597E-04
A10 =	8.56167E-05	-5.45223E-05	-1.35961E-04	-8.16617E-05
A12 =	2.41286E-05			
A14 =	-4.28322E-06			

The equation of the aspheric surface profiles of the fourth embodiment has the same form as that of the first embodiment. Moreover, the definitions of the parameters of the relations are the same as those set forth in the first embodiment, but the value of the relations of the fourth embodiment are listed in the following TABLE 11:

TABLE 11

(Embodiment 4)	
f	5.00
Fno	2.45
HFOV	38.1
V3-V4	38.9
CT5/f	0.12
CT4/CT3	0.29
T23/T12	0.05
R1/f	0.59
R10/f	-0.91
R3/R4	0.24
R6/R5	-1.00
f/f1	-0.94
f/f2	0.36
f/f3- f/f4	0.28
SL/TTL	0.71

Embodiment 5

FIG. 5A shows an image-capturing optical lens assembly in accordance with the fifth embodiment of the present invention, and FIG. 5B shows the aberration curves of the fifth embodiment of the present invention. The image-capturing optical lens assembly in the fifth embodiment mainly comprises five lens elements, in order from an object side to an image side: a front lens group, an aperture stop 500 and a rear lens group, wherein:

the front lens group comprises, in order from the object side to the image side:

a plastic first lens element 510 with negative refractive power having a convex object-side surface 511 and a concave image-side surface 512, both of the surfaces 511 and 512 being aspheric; and

a plastic second lens element 520 with positive refractive power having a convex object-side surface 521 and a concave image-side surface 522, both of the surfaces 521 and 522 being aspheric;

the rear lens group comprises, in order from the object side to the image side:

a glass third lens element 530 with positive refractive power having a convex object-side surface 531 and a convex image-side surface 532;

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a glass fourth lens element **540** with negative refractive power having a concave object-side surface **541** and a concave image-side surface **542**; and

a plastic fifth lens element **550** with positive refractive power having a convex object-side surface **551** and a convex image-side surface **552**, both of the surfaces **551** and **552** being aspheric.

Moreover, the image-capturing optical lens assembly comprises an IR filter **560** disposed between the image-side surface **552** of the fifth lens element **550** and an image plane **571**; the IR filter **560** is made of glass and has no influence on the focal length of the image-capturing optical lens assembly. The image-capturing optical lens assembly further comprises an image sensor **570** provided on the image plane **571**.

The detailed optical data of the fifth embodiment is shown in TABLE 12, and the aspheric surface data is shown in TABLE 13, wherein the units of the curvature radius, the thickness and the focal length are expressed in mm, and HFOV is half of the maximum field of view.

TABLE 12

(Embodiment 5)								
f = 3.36 mm, Fno = 2.00, HFOV = 40.0 deg.								
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal length	
0	Object	Plano	Infinity					
1	Lens 1	5.553200 (ASP)	0.800	Plastic	1.530	55.8	-4.23	
2		1.518430 (ASP)	3.007					
3	Lens 2	2.875220 (ASP)	1.346	Plastic	1.530	55.8	5.75	
4		42.918500 (ASP)	0.218					
5	Ape. Stop	Plano	1.466					
6	Lens 3	4.252	1.751	Glass	1.755	52.3	2.82	
7		-3.509	0.010	Cement	1.514	38.8		
8	Lens 4	-3.509	0.300	Glass	1.847	23.8	-3.37	
9		15.846	0.744					
10	Lens 5	9.260100 (ASP)	1.200	Plastic	1.530	55.8	13.14	
11		-26.816800 (ASP)	1.000					
12	IR-filter	Plano	0.300	Glass	1.516	64.1	—	
13		Plano	1.363					
14	Image	Plano	—					

Reference wavelength is 587.6 nm (d-line)

TABLE 13

Aspheric Coefficients			
Surface #	1	2	3
k =	7.59131E-01	-1.06188E+00	1.02585E+00
A4 =	-3.64147E-03	1.72315E-02	-2.45784E-03
A6 =	-6.26993E-05	-5.42813E-04	9.49002E-04
A8 =	1.82703E-06	4.71397E-04	-3.45011E-04
A10 =	-2.49451E-07	-8.03152E-05	6.80114E-05
Surface #	4	10	11
k =	-1.00000E+00	-9.89213E+00	-1.00000E+00
A4 =	5.62837E-03	-2.63212E-02	-8.95045E-03
A6 =	6.40150E-03	2.20131E-03	-8.75349E-04
A8 =	-2.92462E-03	-2.15527E-03	-1.08664E-04
A10 =	1.16018E-03	3.08103E-04	3.49265E-05

The equation of the aspheric surface profiles of the fifth embodiment has the same form as that of the first embodiment. Moreover, the definitions of the parameters of the relations are the same as those set forth in the first embodiment, but the value of the relations of the fifth embodiment are listed in the following TABLE 14:

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TABLE 14

(Embodiment 5)	
f	3.36
Fno	2.00
HFOV	40.0
V3-V4	28.5
CT5/f	0.36
CT4/CT3	0.17
T23/T12	0.56
R1/f	1.65
R10/f	-7.98
R3/R4	0.07
R6/R5	-0.83
f/f1	-0.79
f/f2	0.58
f/f3- f/f4	0.19
SL/TTL	0.60

Embodiment 6

FIG. 6A shows an image-capturing optical lens assembly in accordance with the sixth embodiment of the present invention, and FIG. 6B shows the aberration curves of the sixth embodiment of the present invention. The image-capturing optical lens assembly in the sixth embodiment mainly comprises five lens elements, in order from an object side to an image side: a front lens group, an aperture stop **600** and a rear lens group, wherein:

the front lens group comprises, in order from the object side to the image side:

a plastic first lens element **610** with negative refractive power having a convex object-side surface **611** and a concave image-side surface **612**, both of the surfaces **611** and **612** being aspheric; and

a plastic second lens element **620** with positive refractive power having a convex object-side surface **621** and a concave image-side surface **622**, both of the surfaces **621** and **622** being aspheric;

the rear lens group comprises, in order from the object side to the image side:

a plastic third lens element **630** with positive refractive power having a convex object-side surface **631** and a convex image-side surface **632**, both of the surfaces **631** and **632** being aspheric;

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a plastic fourth lens element **640** with negative refractive power having a concave object-side surface **641** and a concave image-side surface **642**, both of the surfaces **641** and **642** being aspheric; and

a plastic fifth lens element **650** with positive refractive power having a convex object-side surface **651** and a convex image-side surface **652**, both of the surfaces **651** and **652** being aspheric.

Moreover, the image-capturing optical lens assembly comprises an IR filter **660** disposed between the image-side surface **652** of the fifth lens element **650** and an image plane **671**; the IR filter **660** is made of glass and has no influence on the

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focal length of the image-capturing optical lens assembly. The image-capturing optical lens assembly further comprises an image sensor **670** provided on the image plane **671**, and a stop **680** disposed between the image-side surface **632** of the third lens element **630** and the object-side surface **641** of the fourth lens element **640**.

The detailed optical data of the sixth embodiment is shown in TABLE 15, and the aspheric surface data is shown in TABLE 16, wherein the units of the curvature radius, the thickness and the focal length are expressed in mm, and HFOV is half of the maximum field of view.

TABLE 15

(Embodiment 6)								
f = 4.98 mm, Fno = 2.45, HFOV = 38.1 deg.								
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal length	
0	Object	Plano	Infinity					
1	Lens 1	3.763700 (ASP)	0.500	Plastic	1.544	55.9	-4.29	
2		1.373780 (ASP)	1.566					
3	Lens 2	5.669800 (ASP)	2.147	Plastic	1.634	23.8	11.19	
4		24.070700 (ASP)	0.056					
5	Ape. Stop	Plano	-0.005					
6	Lens 3	6.633900 (ASP)	3.000	Plastic	1.535	56.3	4.35	
7		-3.018700 (ASP)	-0.221					
8	Stop	Plano	0.389					
9	Lens 4	-4.271700 (ASP)	0.300	Plastic	1.634	23.8	-3.87	
10		5.926700 (ASP)	0.897					
11	Lens 5	5.343900 (ASP)	3.000	Plastic	1.544	55.9	5.23	
12		-4.885600 (ASP)	4.000					
13	IR-filter	Plano	0.300	Glass	1.516	64.1	—	
14		Plano	2.883					
15	Image	Plano	—					

Reference wavelength is 587.6 nm (d-line)

Effective radius of Surface 8 is 1.62 mm

TABLE 16

Aspheric Coefficients					
Surface #	1	2	3	4	6
k =	-2.90436E+00	-7.66130E-01	4.49265E+00	-9.90000E+01	-7.75270E+01
A4 =	-8.56799E-03	-6.66963E-03	2.23519E-03	-1.15946E-02	1.08558E-02
A6 =	5.95461E-04	-8.55977E-04	-5.19384E-04	7.63468E-03	-4.77192E-03
A8 =	-1.62992E-05	-1.38695E-04	1.75617E-04	3.11703E-04	4.68839E-03
A10 =		4.44199E-05	-2.05434E-05	-4.32637E-04	-1.20613E-03
A12 =		-1.88126E-07		-2.37429E-13	-2.48709E-13
Surface #	7	9	10	11	12
k =	-5.71612E+00	-1.34646E+01	-3.02246E+01	-1.15361E+01	-2.75210E+00
A4 =	5.89052E-03	5.79392E-03	1.01880E-02	2.98746E-03	-6.06500E-04
A6 =	-1.12077E-02	-1.34059E-02	-6.78877E-03	-9.32537E-05	1.26566E-04
A8 =	1.68886E-03	2.04713E-03	1.67814E-03	1.30239E-05	4.27289E-06
A10 =	-1.09054E-04	-2.12238E-04	-2.29803E-04	-3.38353E-07	7.35260E-07
A12 =	-1.15310E-06	-8.32775E-13	1.32290E-05		

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The equation of the aspheric surface profiles of the sixth embodiment has the same form as that of the first embodiment. Moreover, the definitions of the parameters of the relations are the same as those set forth in the first embodiment, but the value of the relations of the sixth embodiment are listed in the following TABLE 17:

TABLE 17

(Embodiment 6)	
f	4.98
Fno	2.45
HFOV	38.1
V3-V4	32.5
CT5/f	0.60
CT4/CT3	0.10
T23/T12	0.03
R1/f	0.76
R10/f	-0.98
R3/R4	0.24
R6/R5	-0.46
f/f1	-1.16
f/f2	0.45
f/f3- f/f4	-0.14
SL/TTL	0.77

Embodiment 7

FIG. 7A shows an image-capturing optical lens assembly in accordance with the seventh embodiment of the present invention, and FIG. 7B shows the aberration curves of the seventh embodiment of the present invention. The image-capturing optical lens assembly in the seventh embodiment mainly comprises five lens elements, in order from an object side to an image side: a front lens group, an aperture stop **700** and a rear lens group, wherein:

the front lens group comprises, in order from the object side to the image side:

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a plastic first lens element **710** with negative refractive power having a convex object-side surface **711** and a concave image-side surface **712**, both of the surfaces **711** and **712** being aspheric; and

a plastic second lens element **720** with positive refractive power having a convex object-side surface **721** and a concave image-side surface **722**, both of the surfaces **721** and **722** being aspheric;

the rear lens group comprises, in order from the object side to the image side:

a plastic third lens element **730** with positive refractive power having a convex object-side surface **731** and a convex image-side surface **732**, both of the surfaces **731** and **732** being aspheric;

a plastic fourth lens element **740** with negative refractive power having a concave object-side surface **741** and a concave image-side surface **742**, both of the surfaces **741** and **742** being aspheric; and

a plastic fifth lens element **750** with positive refractive power having a convex object-side surface **751** and a convex image-side surface **752**, both of the surfaces **751** and **752** being aspheric.

Moreover, the image-capturing optical lens assembly comprises an IR filter **760** disposed between the image-side surface **752** of the fifth lens element **750** and an image plane **771**; the IR filter **760** is made of glass and has no influence on the focal length of the image-capturing optical lens assembly. The image-capturing optical lens assembly further comprises an image sensor **770** provided on the image plane **771**, and a stop **780** disposed between the image-side surface **732** of the third lens element **730** and the object-side surface **741** of the fourth lens element **740**.

The detailed optical data of the seventh embodiment is shown in TABLE 18, and the aspheric surface data is shown in TABLE 19, wherein the units of the curvature radius, the thickness and the focal length are expressed in mm, and HFOV is half of the maximum field of view.

TABLE 18

(Embodiment 7)								
f = 4.94 mm, Fno = 2.45, HFOV = 35.1 deg.								
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal length	
0	Object	Plano	Infinity					
1	Lens 1	3.989100 (ASP)	0.500	Plastic	1.544	55.9	-4.26	
2		1.402230 (ASP)	1.494					
3	Lens 2	5.616000 (ASP)	2.230	Plastic	1.634	23.8	11.04	
4		24.079200 (ASP)	0.035					
5	Ape. Stop	Plano	-0.005					
6	Lens 3	6.530300 (ASP)	3.000	Plastic	1.535	56.3	4.32	
7		-2.998530 (ASP)	-0.245					
8	Stop	Plano	0.397					
9	Lens 4	-3.815800 (ASP)	0.300	Plastic	1.634	23.8	-3.77	
10		6.579600 (ASP)	0.882					
11	Lens 5	5.010600 (ASP)	3.000	Plastic	1.544	55.9	5.10	
12		-4.900100 (ASP)	4.000					
13	IR-filter	Plano	0.300	Glass	1.516	64.1	—	
14		Plano	2.849					
15	Image	Plano	—					

Reference wavelength is 587.6 nm (d-line)

Effective radius of Surface 8 is 1.645 mm

TABLE 19

Aspheric Coefficients					
Surface #	1	2	3	4	6
k =	-2.81699E+00	-7.99108E-01	4.02921E+00	4.22417E+01	-7.94408E+01
A4 =	-8.38848E-03	-3.98687E-03	2.11172E-03	-1.57199E-02	9.74631E-03

TABLE 19-continued

Aspheric Coefficients					
A6 =	6.50289E-04	-6.98718E-04	-6.40156E-04	8.50150E-03	-5.18712E-03
A8 =	-1.95971E-05	-1.01062E-04	1.54584E-04	-2.17998E-04	4.76370E-03
A10 =		5.42089E-05	-1.62189E-05	-4.37722E-04	-1.31052E-03
A12 =		-1.88383E-07		-2.37402E-13	-2.49024E-13
Surface #	7	9	10	11	12
k =	-7.25028E+00	-1.13850E+01	-4.67010E+01	-1.17351E+01	-3.13327E+00
A4 =	3.54863E-03	7.77099E-03	1.04391E-02	2.57142E-03	-1.03347E-03
A6 =	-1.07124E-02	-1.35244E-02	-6.40507E-03	-5.98586E-05	1.64829E-04
A8 =	1.42456E-03	1.92033E-03	1.61191E-03	1.35890E-05	2.14296E-07
A10 =	-6.58582E-05	-1.67746E-04	-2.25502E-04	-4.31544E-07	9.49148E-07
A12 =	-1.15310E-06	-8.52800E-13	1.32289E-05		

The equation of the aspheric surface profiles of the seventh embodiment has the same form as that of the first embodiment. Moreover, the definitions of the parameters of the relations are the same as those set forth in the first embodiment, but the value of the relations of the seventh embodiment are listed in the following TABLE 20:

TABLE 20

(Embodiment 7)

f	4.94
Fno	2.45
HFOV	35.1
V3-V4	32.5
CT5/f	0.61
CT4/CT3	0.10
T23/T12	0.02
R1/f	0.81
R10/f	-0.99
R3/R4	0.23
R6/R5	-0.46
f/f1	-1.16
f/f2	0.45
f/f3- f/f4	-0.17
SL/TTL	0.77

Embodiment 8

FIG. 8A shows an image-capturing optical lens assembly in accordance with the eighth embodiment of the present invention, and FIG. 8B shows the aberration curves of the eighth embodiment of the present invention. The image-capturing optical lens assembly in the eighth embodiment mainly comprises five lens elements, in order from an object side to an image side: a front lens group, an aperture stop **800** and a rear lens group, wherein:

the front lens group comprises, in order from the object side to the image side:

a plastic first lens element **810** with negative refractive power having a convex object-side surface **811** and a concave image-side surface **812**, both of the surfaces **811** and **812** being aspheric; and

a plastic second lens element **820** with positive refractive power having a convex object-side surface **821** and a concave image-side surface **822**, both of the surfaces **821** and **822** being aspheric;

the rear lens group comprises, in order from the object side to the image side:

a plastic third lens element **830** with positive refractive power having a convex object-side surface **831** and a convex image-side surface **832**, both of the surfaces **831** and **832** being aspheric;

a plastic fourth lens element **840** with negative refractive power having a concave object-side surface **841** and a convex image-side surface **842**, both of the surfaces **841** and **842** being aspheric; and

a plastic fifth lens element **850** with positive refractive power having a convex object-side surface **851** and a convex image-side surface **852**, both of the surfaces **851** and **852** being aspheric.

Moreover, the image-capturing optical lens assembly comprises an IR filter **860** disposed between the image-side surface **852** of the fifth lens element **850** and an image plane **871**; the IR filter **860** is made of glass and has no influence on the focal length of the image-capturing optical lens assembly. The image-capturing optical lens assembly further comprises an image sensor **870** provided on the image plane **871**, and a stop **880** disposed between the image-side surface **832** of the third lens element **830** and the object-side surface **841** of the fourth lens element **840**.

The detailed optical data of the eighth embodiment is shown in TABLE 21, and the aspheric surface data is shown in TABLE 22, wherein the units of the curvature radius, the thickness and the focal length are expressed in mm, and HFOV is half of the maximum field of view.

TABLE 21

(Embodiment 8)							
f = 4.86 mm, Fno = 2.61, HFOV = 35.9 deg.							
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal length
0	Object	Plano	Infinity				
1	Lens 1	4.708600 (ASP)	0.500	Plastic	1.544	55.9	-3.94
2		1.417880 (ASP)	1.462				
3	Lens 2	4.701600 (ASP)	2.620	Plastic	1.634	23.8	11.02
4		11.261600 (ASP)	0.106				
5	Ape. Stop	Plano	-0.005				
6	Lens 3	7.108400 (ASP)	2.800	Plastic	1.535	56.3	4.06

TABLE 21-continued

(Embodiment 8)							
$f = 4.86$ mm, $Fno = 2.61$, $HFOV = 35.9$ deg.							
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal length
7		-2.694900 (ASP)	-0.331				
8	Stop	Plano	0.431				
9	Lens 4	-3.168700 (ASP)	0.311	Plastic	1.650	21.4	-5.41
10		-33.333300 (ASP)	1.571				
11	Lens 5	14.330900 (ASP)	2.628	Plastic	1.544	55.9	7.04
12		-4.894900 (ASP)	4.000				
13	IR-filter	Plano	0.300	Glass	1.516	64.1	—
14		Plano	3.132				
15	Image	Plano	—				

Reference wavelength is 587.6 nm (d-line)

Effective radius of Surface 8 is 1.62 mm

TABLE 22

Aspheric Coefficients					
Surface #	1	2	3	4	6
k =	-1.44609E+00	-7.98147E-01	2.56920E+00	-1.37213E+01	-8.39368E+01
A4 =	-8.28025E-03	-7.46351E-04	-5.04799E-04	-1.63217E-02	4.89906E-03
A6 =	4.52513E-04	-1.33111E-03	-4.36487E-04	4.34370E-03	-6.17631E-03
A8 =	-1.17873E-05	9.57534E-05	1.88280E-04	3.33208E-03	7.36202E-03
A10 =		2.42391E-05	-2.01837E-05	-9.92868E-04	-1.71591E-03
A12 =		-1.32131E-07		-2.24496E-11	-2.24698E-11
Surface #	7	9	10	11	12
k =	-5.24268E+00	-9.39193E+00	-9.84590E+01	-4.03579E+01	-2.80514E+00
A4 =	9.00700E-03	8.56411E-03	8.76857E-03	1.84151E-03	-4.39996E-04
A6 =	-1.06589E-02	-1.39125E-02	-7.41694E-03	-2.00081E-05	1.41960E-04
A8 =	1.04105E-03	1.77615E-03	1.77298E-03	2.07772E-05	-2.79232E-06
A10 =	-6.59139E-05	-2.11608E-04	-2.23783E-04	-7.94085E-07	1.06458E-06
A12 =	-1.15313E-06	-2.31930E-11	1.31740E-05		

The equation of the aspheric surface profiles of the eighth embodiment has the same form as that of the first embodiment. Moreover, the definitions of the parameters of the relations are the same as those set forth in the first embodiment, but the value of the relations of the eighth embodiment are listed in the following TABLE 23:

TABLE 23

(Embodiment 8)	
f	4.86
Fno	2.61
HFOV	35.9
V3-V4	34.9
CT5/f	0.54
CT4/CT3	0.11
T23/T12	0.07
R1/f	0.97
R10/f	-1.01
R3/R4	0.42
R6/R5	-0.38
f/f1	-1.23
f/f2	0.44
f/f3- f/f4	0.30
SL/TTL	0.76

It is to be noted that TABLES 1-23 show different data of the different embodiments; however, the data of the different embodiments are obtained from experiments. Therefore, any image-capturing optical lens assembly of the same structure is considered to be within the scope of the present invention even if it uses different data. The preferred embodiments depicted above are exemplary and are not intended to limit the scope of the present invention.

What is claimed is:

1. An image-capturing optical lens assembly comprising, in order from an object side to an image side, a front lens group, a stop, and a rear lens group, wherein the front lens group comprises, in order from the object side to the image side:
 - a first lens element with negative refractive power having a convex object-side surface and a concave image-side surface;
 - a second lens element with positive refractive power having a convex object-side surface and a concave image-side surface; and
- the rear lens group comprises, in order from the object side to the image side:
 - a third lens element with positive refractive power having a convex object-side surface and a convex image-side surface;
 - a fourth lens element with negative refractive power;
 - a fifth lens element with positive refractive power having a convex image-side surface, and
- wherein a central thickness of the fifth lens element is CT5, a focal length of the lens assembly is f, a curvature radius of the object-side surface of the first lens element is R1, a curvature radius of the image-side surface of the fifth lens element is R10, and they satisfy the following relations:

$$0 < CT5/f < 0.7;$$

$$0 < R1/f < 5.5; \text{ and}$$

$$R10/f < -0.85.$$

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2. The image-capturing optical lens assembly according to claim 1, wherein the fifth lens element is made of plastic and at least one of the object-side and image-side surfaces of the fifth lens element is aspheric.

3. The image-capturing optical lens assembly according to claim 2, wherein an axial distance between the first lens element and the second lens element is T12, an axial distance between the second lens element and the third lens element is T23, and they satisfy the following relation:

$$0 < T23/T12 < 0.65.$$

4. The image-capturing optical lens assembly according to claim 3, wherein the focal length of the lens assembly is f, a focal length of the first lens element is f1, and they satisfy the following relation:

$$-1.6 < f/f1 < -0.6.$$

5. The image-capturing optical lens assembly according to claim 4, wherein the first lens element, the second lens element, the third lens element, the fourth lens element, and the fifth lens element have refractive power.

6. The image-capturing optical lens assembly according to claim 5, wherein an axial distance between the stop and an image plane is SL, an axial distance between the object-side surface of the first lens element and the image plane is TTL, and they satisfy the following relation:

$$0.55 < SL/TTL < 0.85.$$

7. The image-capturing optical lens assembly according to claim 5, wherein an Abbe number of the third lens element is V3, an Abbe number of the fourth lens element is V4, and they satisfy the following relation:

$$28 < V3 - V4 < 45.$$

8. The image-capturing optical lens assembly according to claim 5, wherein the focal length of the lens assembly is f, a focal length of the second lens element is f2, and they satisfy the following relation:

$$0.2 < f/f2 < 0.8.$$

9. The image-capturing optical lens assembly according to claim 5, wherein a curvature radius of the object-side surface of the second lens element is R3, a curvature radius of the image-side surface of the second lens element is R4, and they satisfy the following relation:

$$0 < R3/R4 < 0.5.$$

10. The image-capturing optical lens assembly according to claim 9, wherein the curvature radius of the object-side surface of the third lens element is R5, a curvature radius of

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the image-side surface of the third lens element is R6, and they satisfy the following relation:

$$-2.0 < R6/R5 < -0.2.$$

11. The image-capturing optical lens assembly according to claim 9, wherein the focal length of the lens assembly is f, a focal length of the third lens element is f3, a focal length of the fourth lens element is f4, and they satisfy the following relation:

$$0 < f/f3 - |f/f4| < 0.5.$$

12. The image-capturing optical lens assembly according to claim 5, wherein the third lens element is made of glass.

13. The image-capturing optical lens assembly according to claim 2, wherein an axial distance between the first lens element and the second lens element is T12, an axial distance between the second lens element and the third lens element is T23, and they satisfy the following relation:

$$0 < T23/T12 < 0.2.$$

14. The image-capturing optical lens assembly according to claim 3, wherein the focal length of the lens assembly is f, a focal length of the second lens element is f2, and they satisfy the following relation:

$$0.35 < f/f2 < 0.7.$$

15. The image-capturing optical lens assembly according to claim 3, wherein the fourth lens element has a convex object-side surface and a concave image-side surface.

16. The image-capturing optical lens assembly according to claim 4, wherein half of the maximum field of view of the lens assembly is HFOV, and it satisfies the following relation:

$$35 \text{ degrees} < \text{HFOV} < 45 \text{ degrees}.$$

17. The image-capturing optical lens assembly according to claim 4, wherein a central thickness of the third lens element is CT3, a central thickness of the fourth lens element is CT4, and they satisfy the following relation:

$$0 < CT4/CT3 < 0.3.$$

18. The image-capturing optical lens assembly according to claim 4, wherein the focal length of the lens assembly is f, the focal length of the first lens element is f1, and they satisfy the following relation:

$$-1.3 < f/f1 < -0.9.$$

* * * * *

UNITED STATES PATENT AND TRADEMARK OFFICE
CERTIFICATE OF CORRECTION

PATENT NO. : 8,345,358 B2
APPLICATION NO. : 13/194187
DATED : January 1, 2013
INVENTOR(S) : Chih-Wen Hsu, Ming-Ta Chou and Tsung-Han Tsai

Page 1 of 1

It is certified that error appears in the above-identified patent and that said Letters Patent is hereby corrected as shown below:

Title page, Item 73, line 6: "Largen" should read --Largan--.

Signed and Sealed this
Eighteenth Day of June, 2013

A handwritten signature in cursive script, appearing to read "Teresa Stanek Rea".

Teresa Stanek Rea
Acting Director of the United States Patent and Trademark Office